

TopoSpace

Adaptive Topological State-Space Infrastructure for AI Agents

Technical Whitepaper — Version 2.0, with a controlled simulation study

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Abstract

Contemporary AI agents operate over large, heterogeneous environments composed of prompts, memories, documents, source code, tools, skills, external APIs, other agents, execution histories and continuously changing runtime state. Existing systems organise these objects through vector similarity search, static graphs, hand-written workflows and ever-larger context windows. Such mechanisms answer *which objects are semantically similar?* or *which objects are explicitly connected?* They do not answer the operationally prior question: *which region of the environment is relevant to the agent’s current state and objective?*

We introduce **TopoSpace**, an adaptive topological state-space infrastructure for AI agents. TopoSpace represents AI-addressable entities as points of a parameterised state space and constructs, at each reasoning step, a neighbourhood of the current state using generalised weighted L^p geometry with a task-dependent exponent, task-dependent weights and an adaptive radius, so that the context of an agent in state s pursuing goal g is the sublevel set $N(s, g) = \{x \in \mathcal{X} : d_\theta(s, x) < \varepsilon_\theta(s, g, \mathcal{X})\}$.

This version supplies three things the concept alone does not. First, a complete mathematical development: we prove that weighted L^p aggregation of component pseudometrics is again a pseudometric and characterise exactly when it is a metric; we identify the general class of admissible aggregators (grounded, monotone, subadditive) that preserve every downstream guarantee and show it is closed under conic combination, maximum, concave reshaping and nesting; we prove that weighted power means with $p \geq 1$ are precisely the admissible homogeneous quasi-arithmetic aggregators; we prove exact-recovery, stability, representation, routing, planning, persistence, drift, learning and indexing results, each with the hypotheses it actually requires made explicit. Second, an honest account of what the framework does *not* establish, including the observation that on a finite object set the induced point-set topology is discrete and therefore vacuous, so that the load-bearing topological content lies in the filtration and the drift theory rather than in the core retrieval path. Third, a controlled simulation study: we build a reference simulator with a synthetic multi-view agent environment and report measured results for eleven experiments. Adaptive geometry reaches $F_1 = 0.705 \pm 0.024$ against 0.349 ± 0.013 for the strongest baseline and 0.747 ± 0.010 for an oracle that is told the generator’s hidden structure; the gap radius attains exact recovery on 16.7% of queries against at most 7.1% for any fixed- k rule; routing success peaks at an interior radius (0.645 versus 0.027 under full tool exposure); an ultrametric structural view yields 86.3% exact context reuse against 18.1% for a flat geometry; and every proved bound we could test empirically held on every trial, with the closed-form gradients agreeing with finite differences to a median relative error of 2×10^{-10} .

TopoSpace does not replace embeddings, vector databases, graphs or language models. It introduces an intermediate state and context infrastructure layer that composes those signals into dynamically constructed neighbourhoods, together with

an open-core reference architecture, an SDK surface, an MCP interface, a catalogue of admissible component distances, and a benchmark protocol for testing on real workloads the hypothesis that the simulation supports.

Keywords: AI agents, context engineering, adaptive metrics, L^p geometry, admissible aggregators, ultrametrics, metric learning, topological data analysis, optimal transport, tool routing, agent planning, retrieval infrastructure.

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Part I — Motivation and Foundations

1 Introduction

1.1 From stateless generation to stateful agency

Large language models were first deployed as stateless functions: a prompt entered, a completion left, and nothing persisted. The current generation of systems is different in kind. An agent maintains state across many steps, calls tools, reads and writes files, spawns sub-agents, consults memories written during earlier sessions, and revises its plan in response to observations. In the course of a single task an agent may be nominally able to address its long-term and episodic memories, its documents and knowledge bases, the source files, symbols, dependency edges, tests and commits of a repository, structured databases, external APIs and MCP servers, a registry of tools and skills, other agents with distinct capabilities and costs and trust levels, live execution environments, its own history of actions and errors and outcomes, and the objectives imposed by its user.

The bottleneck has moved. Access is no longer scarce. Nearly everything an agent might need is addressable, and the engineering effort of the past three years has been largely devoted to making it so. What is scarce is *attention*: at any given step the overwhelming majority of addressable objects are irrelevant, and including them is not merely wasteful but actively harmful. It consumes context budget that is paid again at every step of the loop, it dilutes the model’s attention across plausible distractors, and it introduces failure modes — a superseded document confidently quoted, a forbidden tool invoked — that no amount of reasoning capability removes. The question an agent runtime must answer at every step is therefore

what should the agent consider, given where it is and what it is trying to achieve?

(1)

1.2 The thesis

We argue that (1) is best formulated geometrically. Suppose every AI-addressable object, and the agent’s own state, can be compared along a common family of measurements — semantic, temporal, structural, causal, economic, security-related. Then the agent occupies a point s_t of a state space $\mathcal{X}$ and the objects it may consult are other points of the same space, so that context construction becomes the selection of a *neighbourhood* of s_t . The architecture we propose replaces both $\mathcal{X} \rightarrow \text{LLM}$ (push everything into an ever larger window) and $\mathcal{X} \rightarrow \text{Top-}k \rightarrow \text{LLM}$ (push the k semantically nearest objects, for a k fixed offline by a human) with

$$\mathcal{X} \longrightarrow \mathbf{N}(s_t) \longrightarrow \text{LLM}, \quad \mathbf{N}(s_t) \subset \mathcal{X}.$$

Relevance is multi-dimensional. Semantic proximity is one measurement among many. A tool that is semantically apt but forbidden by policy, a memory that is semantically apt but stale, a file that is semantically remote but causally implicated in the current stack trace: all three are mis-ranked by embedding similarity and correctly ranked by an aggregate over several views. Section 4 shows how such an aggregate can be formed without destroying the metric structure that later results consume, and Section 19 measures how much the extra views are worth on a controlled environment where the ground truth is known by construction.

The geometry itself is state-dependent. The exponent p , the weights w and the radius ε are functions of the agent’s state and goal. A debugging step should weight causal and structural evidence heavily and tolerate no violation of the security view, which is a bottleneck geometry with p large; an exploratory research step should weight semantics and permit trade-offs across views, which is a compensatory geometry with p near one. The same object set therefore induces different effective spaces for different agents and different steps. Proposition 4.7 shows this is not a distinction without a difference: the ranking genuinely reverses as p varies, at an exponent we compute in closed form and confirm numerically to six significant figures in Section 19.

The number of retrieved objects is an output, not an input. Fixed Top- k returns exactly k objects whether the truly relevant set has one member or forty, and therefore pays simultaneously in precision and recall. An adaptive radius returns the set that the geometry itself distinguishes. Theorem 7.3 proves that under a margin condition this recovers the relevant set exactly, which no fixed- k rule can do unless k happens to equal the answer’s cardinality; Section 19 measures the effect at 16.7% exact recovery for the gap radius against at most 7.1% for the best fixed k .

1.3 What is claimed, and what is not

We are explicit about epistemic status, because whitepapers in this area routinely blur the boundary between theorem, measurement and aspiration.

Proved. Every statement labelled Theorem, Proposition, Lemma or Corollary is proved, in the main text or in Appendix A. These are mathematical facts about the construction and hold independently of any empirical outcome.

Measured. Part V reports a controlled simulation. The environment is synthetic and its generator is described completely in Section 18 and Appendix C; the code is part of the open-source distribution and the reported numbers are reproducible from a fixed seed in under a minute on one core. Simulation establishes that the mechanisms behave as the theory predicts and quantifies how much each ingredient contributes when the multi-view hypothesis is true. It does not establish that real agent workloads satisfy that hypothesis. We say so again in Section 20 and design the field protocol of Section 25 to test precisely that.

Hypothesised. That adaptive topological neighbourhoods outperform embedding Top- k , hybrid retrieval, graph retrieval and full-context exposure on *real* agent tasks remains a hypothesis, stated with its falsification conditions in Section 25.

Deliberately deflated. We record where topological language is load-bearing and where it is merely suggestive. Remark 4.9 observes that on a finite object set the metric topology is discrete and therefore vacuous by itself; the non-trivial topological content lies in the filtration of Section 8 and the drift theory of Section 15, not in the point-set topology of $\mathcal{X}$. Saying so is more useful than borrowing the prestige of the word.

1.4 Organisation

Part I motivates the problem, reviews the limitations of existing context infrastructure and formalises the state model. Part II develops the mathematics: weighted L^p ag-

gregation, the general theory of admissible aggregators with a catalogue of component distances, adaptive geometry, adaptive radii, multi-scale filtrations, and an axiomatic characterisation of context selection. Part III applies the theory to memory, tool and skill routing, multi-agent orchestration, planning, coding agents and drift detection. Part IV covers learning the geometry and the computational cost of doing so. Part V describes the reference simulator, states the eleven experiments and reports their measured results. Part VI gives the system and open-source architecture, the developer and MCP interfaces, the field benchmark protocol, the limitations, the open research questions and the positioning. Appendix A contains deferred proofs, Appendix B the distance catalogue, Appendix C the simulator specification, and Appendix D a notation table.

2 Limitations of Existing Context Infrastructure

This section states as precisely as we can what the incumbent mechanisms do and do not accomplish. The point is not that they are wrong. Each is excellent at the job it was designed for, and Proposition 4.10 shows each is a point of the TopoSpace hypothesis class. The point is that none of them is a formulation of (1).

2.1 Vector retrieval

A retrieval-augmented system embeds a query q as $E(q) \in \mathbb{R}^d$ and returns

$$\text{Top}_k(q) = \arg \max_{S \subseteq \mathcal{X}, |S|=k} \sum_{x \in S} \text{sim}(E(q), E(x)), \quad (2)$$

typically with sim the cosine similarity and the maximisation performed approximately by an ANN index [23, 17]. This is the backbone of retrieval-augmented generation [21] and it works: for “what did the document say about X ”, semantic proximity is very nearly the right notion of relevance.

Relevance is projected onto a single axis. Equation (2) compresses every consideration into one scalar. Operational relevance for an agent depends jointly on semantics, on dependency (does this symbol appear in the failing call stack?), on recency (is this memory still valid?), on provenance, on cost and latency (is this tool affordable at this step?), on risk and permission (is this action allowed for this principal?), on reliability (has this tool failed twice already?) and on causality (did this action produce the current error?). Semantic proximity neither implies nor is implied by operational relevance, and the counterexamples are not exotic: the stack-trace file that shares no vocabulary with the request, the deprecated API document that is the single best semantic match in the corpus. Our simulator injects exactly these two populations — semantically near decoys and semantically remote positives — and Section 19 reports how sharply Top_k degrades as their number grows while the multi-view geometry does not.

The cardinality is exogenous. The parameter k is chosen by an engineer, offline, once, and applied to every query, while the truly relevant set has variable cardinality. Fixed k therefore pays in precision when the relevant set is smaller and in recall when it is larger. Section 7 makes this exact and Section 19 measures it.

Similarity is not a metric. Cosine similarity is not a distance, and the cosine dissimilarity $1 - \cos$ violates the triangle inequality. This is harmless for ranking but fatal for any argument that composes distances, as planning, filtration and drift analysis all do. TopoSpace therefore works with the angular distance, a genuine metric on the sphere, or with normalised Euclidean distance, and states its metric hypotheses explicitly rather than assuming them.

2.2 Static graphs

Graph-structured retrieval [12] represents relations explicitly as $G = (V, E)$ and traverses them. Where the relation is known a priori and stable — imports, citations, foreign keys, call edges — this is strictly more informative than embedding proximity. The limitation is that the *relevant* relation is task-dependent. Two files related only by “both were touched in the commit that introduced the regression” are strongly related while debugging that regression and unrelated afterwards, and a static edge set must either omit such edges, losing recall, or include all of them, in which case traversal degenerates because almost everything is reachable within two hops. TopoSpace treats relatedness as state-dependent: the effective edge set induced by $N(\cdot)$ changes when (s, g) changes without any rewrite of the underlying store. Formally a static graph is the special case in which the weights and the radius are constant functions.

2.3 Large context windows

Context windows have grown by orders of magnitude, which tempts the architecture $\mathcal{X} \rightarrow \text{LLM}$. Three objections apply. Cost and latency scale with the material supplied, and in an agent loop that cost is paid at every step rather than once. The empirical literature on long-context behaviour reports degradation when relevant material is surrounded by plausible irrelevant material [22, 31], so a larger window enlarges the haystack as well as the space for the needle. Most fundamentally, a window is a capacity rather than a decision procedure: it permits the model to inspect more, but does not determine what deserves attention. Selection remains necessary, and if it is necessary it should be performed by a component designed for it rather than implicitly by attention. Our routing experiment quantifies the cost of declining to select: with a selection-accuracy curve calibrated to degrade mildly in the number of exposed candidates, full exposure of six hundred tools yields a success probability of 0.027 against 0.645 at the optimal radius.

2.4 Hand-written orchestration

Explicit workflows — “after a test failure read the failing test, then the file under test, then the last three commits” — encode exactly the neighbourhood intuition this whitepaper formalises, but as static code. They are precise, auditable and brittle: each new situation requires a new branch, and the branches interact. TopoSpace can be read as an attempt to replace a combinatorial explosion of hand-written selection rules with one parameterised selection rule whose parameters are configured or learned, at the cost of making the parameters themselves an object of study.

Mechanism	Question answered	Structural limitation for agents
Vector retrieval	What is semantically similar?	One axis; exogenous k ; similarity is not a metric
Knowledge graph	What is explicitly connected?	Relation fixed offline; task-dependence lost
Large context	(capacity, not a question)	Cost per step; distractor degradation; no selection
Hand-written flow	What did the engineer foresee?	Combinatorial branch growth; brittle
TopoSpace	What lies in the relevant neighbourhood of the current state?	Requires calibration of the views, the aggregator, the weights and the radius

Table 1: Positioning of TopoSpace relative to incumbent context mechanisms.

3 The TopoSpace State Model

3.1 Objects

Definition 3.1 (Object universe). The *object universe* is a finite set $\mathcal{X} = M \cup K \cup T \cup S \cup A \cup C \cup H$ whose constituents are, respectively, memories, knowledge objects, tools, skills, agents, code and computational objects, and historical states and actions. The constituents need not be disjoint: a skill may also be a knowledge object and a past action may also be a memory. We write $N = |\mathcal{X}|$.

Why one union rather than seven stores. The union is the first substantive modelling decision. Conventional stacks segregate these types — a vector database for memories, a registry for tools, a configuration file for agents, a repository index for code — and therefore cannot express a comparison across them. Yet agents make exactly such comparisons: *should I recall what happened last time, call the search tool, or delegate to the analysis agent?* Placing all types in one space is what makes that question expressible as a single neighbourhood query, and it costs nothing at the storage layer, since Section 21 keeps the physical stores where they are.

3.2 From coordinates to component dissimilarities

The natural first attempt is to give every object a coordinate vector $\phi(x) \in \mathbb{R}^n$ and compare coordinates. This is adequate for numeric attributes such as cost, latency, timestamp or risk score, and inadequate for the attributes that matter most. Semantic position is a high-dimensional embedding rather than a scalar; structural position in a dependency graph has no canonical coordinates at all; causal relation is intrinsically a relation between two objects rather than a property of one. We therefore work with component dissimilarities, which are both more general and better behaved.

Definition 3.2 (View system). A *view system* on $\mathcal{X}$ is a finite family $\delta = (\delta_1, \dots, \delta_n)$ of functions $\delta_i : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$. The view δ_i is *metric* if it is a pseudometric, that is $\delta_i(x, x) = 0$, $\delta_i(x, y) = \delta_i(y, x)$ and $\delta_i(x, z) \leq \delta_i(x, y) + \delta_i(y, z)$; *quasi-metric* if symmetry may fail while the remaining axioms hold; and *raw* otherwise.

Coordinates are the special case $\delta_i(x, y) = |\phi_i(x) - \phi_i(y)|$, which is always a pseudo-metric, so nothing is lost by the generalisation. Table 2 lists the canonical views of the reference implementation; Appendix B catalogues the alternatives.

View	Symbol	Construction	Class
Semantic	δ_{sem}	angular distance $\frac{2}{\pi} \arccos(\langle E(x), E(y) \rangle / \ E(x)\ \ E(y)\)$	metric
Temporal	δ_{tim}	$\psi(t(x) - t(y))$ with ψ as in Lemma 3.3	metric
Structural	δ_{str}	normalised shortest-path length in the dependency graph	metric
Causal	δ_{cau}	1 – estimated probability that x lies on a causal path to y	quasi-metric
Cost	δ_{cst}	$ c(x) - c(y) $ on a normalised cost scale	metric
Risk	δ_{rsk}	$ r(x) - r(y) $ with $r \in [0, 1]$ a policy risk score	metric
Reliability	δ_{rel}	$ \rho(x) - \rho(y) $ with ρ an exponentially weighted success rate	metric
Security	δ_{sec}	0 if the permission sets are compatible, 1 otherwise	metric
Task	δ_{tsk}	normalised Hamming or Jaccard distance between task-tag sets	metric
History	δ_{his}	1 – co-occurrence rate in past successful trajectories	pseudometric [†]

Table 2: Canonical views. [†] The history view is a pseudometric only after the correction of Lemma 3.4; see Remark 3.5.

Lemma 3.3 (Concave reparameterisation preserves metricity). *Let δ be a pseudometric on $\mathcal{X}$ and let $\psi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be non-decreasing, concave and satisfy $\psi(0) = 0$. Then $\psi \circ \delta$ is a pseudometric. In particular $\psi(t) = t/(1+t)$, $\psi(t) = \min(t, 1)$ and $\psi(t) = 1 - e^{-\lambda t}$ produce pseudometrics with values in a bounded interval.*

Proof. Non-negativity, $\psi(\delta(x, x)) = \psi(0) = 0$ and symmetry are immediate. A concave ψ with $\psi(0) = 0$ is subadditive: for $a, b \geq 0$ with $a + b > 0$, concavity applied to the convex combination $a = \frac{a}{a+b}(a+b) + \frac{b}{a+b} \cdot 0$ gives $\psi(a) \geq \frac{a}{a+b}\psi(a+b)$, symmetrically $\psi(b) \geq \frac{b}{a+b}\psi(a+b)$, and adding yields $\psi(a) + \psi(b) \geq \psi(a+b)$. Hence, by monotonicity and subadditivity, $\psi(\delta(x, z)) \leq \psi(\delta(x, y) + \delta(y, z)) \leq \psi(\delta(x, y)) + \psi(\delta(y, z))$. $\square$

Lemma 3.3 is the workhorse of the normalisation layer. It lets us map every view onto a common bounded scale — without which weighted aggregation across views is meaningless, since a weight multiplies a quantity whose units would otherwise be arbitrary — while guaranteeing that we have not destroyed the metric structure the later theorems consume. It is also the reason the reference implementation exposes normalisation as a declarative choice among a small set of admissible ψ rather than as free-form user code.

Lemma 3.4 (Metrisation of a similarity). *Let $\kappa : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ be symmetric with $\kappa(x, x) = 1$. Then $\delta(x, y) = 1 - \kappa(x, y)$ need not satisfy the triangle inequality; however, if κ is a positive semidefinite kernel then $\delta_{\mathcal{H}}(x, y) = \sqrt{2 - 2\kappa(x, y)}$ is a pseudometric, being the distance in the reproducing-kernel Hilbert space of κ .*

Proof. Let Φ be the canonical feature map, so that $\kappa(x, y) = \langle \Phi(x), \Phi(y) \rangle_{\mathcal{H}}$ and $\|\Phi(x)\|^2 = 1$. Then $\|\Phi(x) - \Phi(y)\|^2 = 2 - 2\kappa(x, y)$, so $\delta_{\mathcal{H}}$ is the pullback of a Hilbert-space norm distance and inherits its axioms. $\square$

Remark 3.5 (Co-occurrence views). Raw co-occurrence rates are similarities rather than distances, and $1 - \kappa$ violates the triangle inequality in general. The implementation therefore projects the empirical co-occurrence matrix onto the positive semidefinite cone and applies Lemma 3.4 before the view is admitted. Where the projection is too expensive the view is declared raw, and the corresponding guarantees downgrade in the manner tabulated in Table 4 — which is enough for ranking and neighbourhood construction but not for the planning bounds of Section 13.

3.3 States and goals

Definition 3.6 (State and goal). A *state* $s \in \mathcal{S}$ is an object of the same kind as the elements of $\mathcal{X}$, in the sense that the views extend to $\delta_i(s, x)$ for every $x \in \mathcal{X}$. A *goal* $g \in \mathcal{G}$ is a specification of the objective, itself embedded so that goal-conditioned quantities are defined. The reference implementation represents the state of a coding agent as the tuple of its goal, its open code, its runtime, its outstanding errors, its test results, its history and its constraints, each field contributing to one or more views.

Why the agent is a point of its own world. Treating the state as a point of the same space as the objects is the second substantive modelling decision. It converts “retrieve documents similar to the query” into “find the region of the world near where I am standing”, and it is what allows history, errors and constraints — properties of the agent rather than of any document — to participate in the comparison at all. It also explains why the resulting dissimilarity is directional (Proposition 6.2): the stack trace is relevant to the debugging agent, while the debugging agent is not relevant to the stack trace in the same sense.

Remark 3.7 (Cold start). Views that depend on history are undefined at first contact with a new object. The implementation initialises them at the population mean and holds the corresponding weight at zero until a minimum observation count is reached, which by Theorem 4.3 leaves the aggregate a valid pseudometric throughout, since zero weights are permitted and merely remove the view from the support.

Part II — The Mathematics of Adaptive Neighbourhoods

4 Weighted L^p Geometry

4.1 Definition and interpretation

Definition 4.1 (Weighted L^p aggregation). Let $\delta = (\delta_1, \dots, \delta_n)$ be a view system on $\mathcal{X}$, let $p \in [1, \infty]$ and let $w \in \mathbb{R}_{\geq 0}^n$ be a weight vector normalised so that $\sum_i w_i = 1$ unless stated otherwise. The *weighted L^p aggregate* is

$$d_{p,w}(x, y) = \begin{cases} \left(\sum_{i=1}^n w_i \delta_i(x, y)^p \right)^{1/p}, & 1 \leq p < \infty, \\ \max_{i \in \text{supp}(w)} \delta_i(x, y), & p = \infty, \end{cases} \quad (3)$$

where $\text{supp}(w) = \{i : w_i > 0\}$.

The exponent is a semantic control. Three readings of (3) explain why p is a meaningful parameter rather than an opaque tuning knob. At $p = 1$ the aggregation is *compensatory*: a large discrepancy on one view can be offset by small discrepancies on the others, which is right when the views are substitutable pieces of evidence. At $p = \infty$ it is *non-compensatory*, or bottleneck: the aggregate equals the worst view, so no amount of semantic similarity rescues an object that violates the security view, which is right when some views encode hard constraints. Intermediate exponents interpolate, and Theorem 4.6 shows the interpolation is monotone, so increasing p moves continuously from average-case to worst-case aggregation. The choice between the two regimes is a genuine modelling decision that differs across tasks and even across steps of one task, and that observation is the entire motivation for making p adaptive.

4.2 Metric properties

Lemma 4.2 (Minkowski’s inequality, weighted form). *Let $p \in [1, \infty)$, $w \in \mathbb{R}_{\geq 0}^n$ and $a, b \in \mathbb{R}_{\geq 0}^n$. Then $(\sum_i w_i (a_i + b_i)^p)^{1/p} \leq (\sum_i w_i a_i^p)^{1/p} + (\sum_i w_i b_i^p)^{1/p}$.*

Proof. Appendix A.1, by the standard derivation from Hölder’s inequality applied to the measure assigning mass w_i to coordinate i . $\square$

Theorem 4.3 (Aggregation theorem). *Let each view δ_i be a pseudometric on $\mathcal{X}$, let $p \in [1, \infty]$ and $w \in \mathbb{R}_{\geq 0}^n$. Then $d_{p,w}$ defined by (3) is a pseudometric on $\mathcal{X}$. It is a metric if and only if the family $\{\delta_i\}_{i \in \text{supp}(w)}$ separates points, meaning that for all $x \neq y$ there is $i \in \text{supp}(w)$ with $\delta_i(x, y) > 0$. If some views are quasi-metrics and the rest are pseudometrics then $d_{p,w}$ is a quasi-pseudometric. If some views are merely non-negative and vanish on the diagonal then $d_{p,w}$ is a non-negative dissimilarity vanishing on the diagonal, and no triangle inequality is claimed.*

Proof. Non-negativity is clear and $d_{p,w}(x, x) = 0$ because every $\delta_i(x, x) = 0$; symmetry holds coordinatewise and is preserved by (3). For the triangle inequality fix x, y, z and set $a_i = \delta_i(x, y)$, $b_i = \delta_i(y, z)$, $c_i = \delta_i(x, z)$, so that $c_i \leq a_i + b_i$ for every i . For $p < \infty$ the map $u \mapsto (\sum_i w_i u_i^p)^{1/p}$ is non-decreasing in each non-negative coordinate, hence

$$d_{p,w}(x, z) = \left(\sum_i w_i c_i^p \right)^{1/p} \leq \left(\sum_i w_i (a_i + b_i)^p \right)^{1/p} \leq \left(\sum_i w_i a_i^p \right)^{1/p} + \left(\sum_i w_i b_i^p \right)^{1/p},$$

the last step by Lemma 4.2; for $p = \infty$ the chain $\max_i c_i \leq \max_i (a_i + b_i) \leq \max_i a_i + \max_i b_i$ gives the same conclusion. If the family separates points then $x \neq y$ forces $\delta_i(x, y) > 0$ for some i with $w_i > 0$, so $d_{p,w}(x, y) > 0$; conversely if $\delta_i(x, y) = 0$ for all $i \in \text{supp}(w)$ with $x \neq y$ then $d_{p,w}(x, y) = 0$. The quasi-metric claim follows because the triangle argument nowhere used symmetry, and the last claim because that argument required $c_i \leq a_i + b_i$. $\square$

What the theorem licenses, and what it forbids. Theorem 4.3 is the licence for the entire architecture: it says heterogeneous notions of relatedness may be combined into one distance without losing the axioms that the planning bounds, the filtration stability and the indexing results consume, provided each component is itself well behaved and $p \geq 1$. It also identifies precisely which engineering decision is dangerous. Admitting a raw similarity as a view — the last case of the theorem — silently downgrades every downstream guarantee, and does so without any visible symptom at the retrieval layer, since ranking continues to work. This is why the reference runtime records the class of every admitted view and can report which theorems still apply to a given configuration.

Corollary 4.4 (Metric quotient). *Let $d_{p,w}$ be a pseudometric on $\mathcal{X}$ and define $x \sim y$ iff $d_{p,w}(x, y) = 0$. Then $\sim$ is an equivalence relation and $d_{p,w}$ descends to a metric on $\mathcal{X}/\sim$.*

Proof. Reflexivity and symmetry are clear, and transitivity follows from the triangle inequality since $d_{p,w}(x, y) = d_{p,w}(y, z) = 0$ forces $d_{p,w}(x, z) \leq 0$. If $x \sim x'$ and $y \sim y'$ then $|d_{p,w}(x, y) - d_{p,w}(x', y')| \leq d_{p,w}(x, x') + d_{p,w}(y, y') = 0$, so the induced map on classes is well defined, and by construction it vanishes only on the diagonal. $\square$

Corollary 4.4 has a concrete operational meaning that is easy to miss. Objects at distance zero under the currently active weighting are, for the purposes of this step, indistinguishable, so collapsing them is not information loss but deduplication. The reference implementation exploits this to suppress near-duplicate memories and redundant tool descriptions whenever the active weights ignore the dimensions along which they differ — a behaviour that changes automatically with the task, since the quotient is taken with respect to the current geometry rather than a fixed one.

4.3 Why the exponent may not fall below one

Proposition 4.5 (Failure below $p = 1$). *Let $0 < p < 1$. Then (3) need not satisfy the triangle inequality: with $n = 2$, $w = (1, 1)$ and three objects such that $\delta(x, y) = (1, 0)$, $\delta(y, z) = (0, 1)$ and $\delta(x, z) = (1, 1)$, one has $d_{p,w}(x, y) = d_{p,w}(y, z) = 1$ while $d_{p,w}(x, z) = 2^{1/p} > 2$. Conversely the unaggregated power sum $\rho_{p,w}(x, y) = \sum_i w_i \delta_i(x, y)^p$ is a pseudometric for $0 < p \leq 1$.*

Proof. The counterexample is a computation: for $0 < p < 1$ we have $1/p > 1$ and hence $2^{1/p} > 2 = d_{p,w}(x, y) + d_{p,w}(y, z)$. For the converse, $t \mapsto t^p$ is concave on $\mathbb{R}_{\geq 0}$ with value 0 at 0 and hence subadditive by the argument of Lemma 3.3, so $\delta_i(x, z)^p \leq (\delta_i(x, y) + \delta_i(y, z))^p \leq \delta_i(x, y)^p + \delta_i(y, z)^p$, and summation against non-negative weights preserves the inequality. $\square$

Proposition 4.5 explains a design constraint that recurs in Section 16: the learned exponent must be confined to $[1, \infty)$, which we enforce by the parameterisation $p_\theta = 1 + \text{softplus}(\cdot)$ rather than by a penalty that could be violated transiently during optimisation. Sub-linear aggregation is not forbidden in principle, since $\rho_{p,w}$ remains a pseudometric, but it is a different object and the two must not be conflated.

4.4 The exponent family is a monotone band

Theorem 4.6 (Power-mean monotonicity and sandwich). *Let w satisfy $w_i \geq 0$ and $\sum_i w_i = 1$, and let $a \in \mathbb{R}_{\geq 0}^n$ collect the view values $a_i = \delta_i(x, y)$. Write $M_p(a) = (\sum_i w_i a_i^p)^{1/p}$ for $p \in [1, \infty)$ and $M_\infty(a) = \max_{i \in \text{supp}(w)} a_i$. Then $p \mapsto M_p(a)$ is non-decreasing on $[1, \infty]$; consequently $M_1(a) \leq M_p(a) \leq M_\infty(a)$ for every p , and $|M_p(a) - M_q(a)| \leq M_\infty(a) - M_1(a)$ for all $p, q \in [1, \infty]$; finally $M_p(a) \rightarrow M_\infty(a)$ as $p \rightarrow \infty$.*

Proof. For $1 \leq p < q < \infty$ put $u_i = a_i^p$; the function $t \mapsto t^{q/p}$ is convex on $\mathbb{R}_{\geq 0}$ because $q/p > 1$, so Jensen's inequality with the probability weights w gives $(\sum_i w_i u_i)^{q/p} \leq \sum_i w_i u_i^{q/p} = \sum_i w_i a_i^q$, and raising to the power $1/q$ yields $M_p(a) \leq M_q(a)$. For the limit let $m = \max_{i \in \text{supp}(w)} a_i$ attained at j ; then $\sum_i w_i a_i^p \geq w_j m^p$ and $\sum_i w_i a_i^p \leq m^p$, so $w_j^{1/p} m \leq M_p(a) \leq m$ and $w_j^{1/p} \rightarrow 1$. The sandwich and the band width follow. $\square$

The exponent matters exactly when the views disagree. Theorem 4.6 clarifies what varying p can and cannot do. It cannot make distances arbitrary: every member of the family lies in the band $[M_1, M_\infty]$, whose width measures how much the views disagree. When all views agree the band collapses and p is irrelevant. This is a satisfying formal counterpart of the intuition that multi-criteria aggregation is only interesting under conflict, and it also predicts where adaptivity will fail to pay: in a deployment whose views are highly correlated, learning p buys nothing, and the practitioner should discover this from the band width rather than from a disappointing benchmark.

4.5 Adaptivity in the exponent changes the ranking

Monotonicity might suggest that p merely rescales distances and leaves the induced ordering — the only thing a neighbourhood query depends on — unchanged. That is false, and the following proposition is the precise reason the framework needs an adaptive exponent at all.

Proposition 4.7 (Rank crossover). *Let $w = (\frac{1}{2}, \frac{1}{2})$ and consider two candidates with view vectors $a = \delta(s, x) = (1, 0)$ and $b = \delta(s, y) = (0.6, 0.6)$. Then $M_1(a) = 0.5 < 0.6 = M_1(b)$, so x is nearer at $p = 1$, while $M_\infty(a) = 1 > 0.6 = M_\infty(b)$, so y is nearer at $p = \infty$. Since $M_p(a) = 2^{-1/p}$ and $M_p(b) = 0.6$ for all p , the ranking reverses at the unique crossover exponent $p^* = \ln(1/2)/\ln(0.6) = 1.3569\dots$. More generally, if two candidates satisfy $M_1(a) < M_1(b)$ and $M_\infty(a) > M_\infty(b)$ then there exists $p^* \in (1, \infty)$ at which the induced ranking changes.*

Proof. For $a = (1, 0)$ we have $M_p(a) = (\frac{1}{2} \cdot 1^p + \frac{1}{2} \cdot 0^p)^{1/p} = 2^{-1/p}$, and for $b = (0.6, 0.6)$ we have $M_p(b) = 0.6$ independently of p . Setting $2^{-1/p} = 0.6$ gives $-\frac{1}{p} \ln 2 = \ln 0.6$, that is $p^* = \ln(1/2)/\ln(0.6) \approx 1.357$; since $p \mapsto 2^{-1/p}$ is strictly increasing the crossing is unique, with $M_p(a) < M_p(b)$ below p^* and $M_p(a) > M_p(b)$ above it, consistent with $M_1(a) = 0.5$ and $M_2(a) = 2^{-1/2} \approx 0.707$. For the general statement, $f(p) = M_p(a) - M_p(b)$ is continuous on $[1, \infty)$ and extends continuously to $p = \infty$ by Theorem 4.6; since $f(1) < 0$ and $\lim_{p \rightarrow \infty} f(p) > 0$, the intermediate value theorem supplies a finite zero. $\square$

The example rewards an operational reading. Candidate x is a perfect match on one view and a total mismatch on the other, for instance a tool that is exactly right semantically and wrong on the security view; candidate y is mediocre on both. A compensatory agent prefers x and a constraint-respecting agent prefers y , and neither preference is universally correct, which is why p must be a function of the state rather than a constant compiled into the retrieval layer. Section 19 confirms the crossover numerically to six significant figures and reports that, on our synthetic environment, 19.0% of candidate pairs reverse their order between $p = 1$ and $p = 8$ while the top-ten sets differ by a Jaccard distance of 0.740; the exponent is not a cosmetic parameter.

4.6 Relation to Mahalanobis metric learning

Proposition 4.8 (Strict generalisation of diagonal Mahalanobis). *Let $\delta_i(x, y) = |\phi_i(x) - \phi_i(y)|$ be coordinate views. Then $d_{2,w}$ coincides with the Mahalanobis metric induced by $A = \text{diag}(w)$, while for $p \neq 2$ the aggregate $d_{p,w}$ is not induced by any inner product and hence is not a Mahalanobis metric for any positive semidefinite A .*

Proof. The first claim is immediate from $d_A(x, y)^2 = (x - y)^\top A(x - y) = \sum_i w_i (x_i - y_i)^2$. For the second, a metric induced by an inner product satisfies the parallelogram law $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$; taking $n = 2$, $w = (1, 1)$, $u = (1, 0)$, $v = (0, 1)$ and

$p = 1$ gives $4 + 4 = 8$ on the left and $2 + 2 = 4$ on the right, and the same computation at $p = \infty$ gives $1 + 1 = 2$ against 4. For general $p \neq 2$ the failure follows from the classical fact that ℓ_p^n with $n \geq 2$ is a Hilbert space only for $p = 2$. $\square$

Metric learning in the Mahalanobis family [34] is thus the $p = 2$ slice of the TopoSpace hypothesis class. The additional freedom in p purchases exactly the compensatory versus non-compensatory axis discussed above, which no reweighting of a Euclidean metric can express, and which Section 19 finds the fitting procedure selecting away from $p = 2$ on three of four task types.

4.7 The role of topology, stated without inflation

Remark 4.9 (On finite object sets the metric topology is discrete). If $d_{p,w}$ is a metric and $\mathcal{X}$ is finite then every singleton is open, taking ε below the minimum pairwise distance, so the induced topology is discrete and carries no information. Moreover all norms on $\mathbb{R}^n$ are equivalent, so for fixed strictly positive weights every $p \in [1, \infty]$ induces the same topology; by Theorem 4.6 the family is even uniformly comparable. The value of varying p and w therefore lies entirely in the induced ranking and in the shape of balls at a finite scale, not in the point-set topology. The genuinely topological content of TopoSpace appears in two later places: the filtration $\{B_\varepsilon\}_{\varepsilon>0}$ of Section 8, whose homology is a non-trivial invariant of the scale parameter even for finite $\mathcal{X}$, and the drift theory of Section 15, which compares environments by comparing those invariants. We flag this explicitly rather than let the name of the system do argumentative work it has not earned.

4.8 Existing mechanisms as special cases

Proposition 4.10 (Generalisation). *The following are instances of Definition 4.1 together with a radius rule. Embedding Top- k is the case $n = 1$ with δ_1 the angular distance and ε the k -th smallest value of $\delta_1(s, \cdot)$, as proved in Proposition 7.2. One-hop graph retrieval is $n = 1$ with δ_1 the graph shortest-path distance and $\varepsilon = 1.5$, and m -hop retrieval is $\varepsilon = m + \frac{1}{2}$. Hybrid retrieval with fixed mixing λ is $n = 2$, $p = 1$, $w = (\lambda, 1 - \lambda)$. Full-context exposure is $\varepsilon = \infty$. Rule-based filtering followed by ranking is obtained by admitting the filter predicate as a $\{0, 1\}$ -valued view with $p = \infty$, which places every violating object strictly after every compliant one provided the remaining views are normalised below 1.*

Proof. Each is a substitution into (3) and the definition of a ball. The first is proved as Proposition 7.2; the last uses $M_\infty = \max_i a_i \geq 1$ whenever the filter view equals 1. $\square$

The practical import is that TopoSpace can be deployed incrementally. An installation may begin in the configuration that reproduces its existing retrieval stack exactly, then introduce additional views and adaptivity one at a time, with the guarantee that the baseline remains inside the hypothesis class and can be recovered by setting weights to zero. It also means that a loss to a baseline in evaluation is informative rather than confusing: it indicates a failure of the fitting procedure to find a point of its own search space, not a mismatch between frameworks.

4.9 High-dimensional concentration and the case for sparse weights

Proposition 4.11 (Distance concentration). *Fix $p \in [1, \infty)$ and suppose that for a random object X the view values $\delta_i(s, X)^p$ are independent across i with common mean $\mu > 0$ and variance $\sigma^2 < \infty$. Let w be a probability weight vector and $S = \sum_i w_i \delta_i(s, X)^p$. Then*

$\mathbb{E}[S] = \mu$ and $\text{Var}(S) = \sigma^2 \|w\|_2^2 = \sigma^2/n_{\text{eff}}$ where $n_{\text{eff}} = \|w\|_2^{-2} \in [1, n]$, so the relative standard deviation of S is $\sigma/(\mu\sqrt{n_{\text{eff}}})$ and decays as $n^{-1/2}$ under uniform weighting.

Proof. Linearity gives $\mathbb{E}[S] = \mu \sum_i w_i = \mu$ and independence gives $\text{Var}(S) = \sum_i w_i^2 \sigma^2$. For $w_i = 1/n$ we have $\|w\|_2^2 = 1/n$; the bounds on n_{eff} follow from $1/n \leq \|w\|_2^2 \leq 1$ for probability vectors, with the extremes at the uniform and one-hot weightings. $\square$

Proposition 4.11 formalises the phenomenon reported empirically by Beyer et al. [2] and analysed by Aggarwal et al. [1]: as the number of pooled dimensions grows, all distances converge to a common value and the contrast between near and far objects vanishes, degrading any neighbourhood query. Two design consequences follow. Sparse weights are a statistical necessity rather than an interpretability nicety, because the participation ratio n_{eff} rather than the raw view count governs concentration, which motivates the entropy regularisation of Section 16. And lower exponents are safer in high dimension, since contrast degrades faster for larger p , so the useful range of p is dimension-dependent as well as state-dependent. Section 19 measures both effects: relative contrast falls from 370 at $n = 2$ to 0.34 at $n = 128$ under uniform weights, while a weighting concentrated on three views holds contrast above 17 throughout.

5 Admissible Aggregators and a Catalogue of Geometries

Section 4 fixed one aggregation rule and left the component views abstract. Both choices deserve scrutiny. This section identifies the exact property an aggregation rule must have for Theorem 4.3 to survive, shows that L^p is canonical within a natural class without being the only admissible choice, records which component distances suit which kind of AI object, and states precisely which downstream guarantees are lost when a component fails to be a metric. The full catalogue is Appendix B; this section supplies the theory that tells a practitioner how to read it.

5.1 Admissible aggregators

Definition 5.1 (Admissible aggregator). A function $A : \mathbb{R}_{\geq 0}^n \rightarrow \mathbb{R}_{\geq 0}$ is an *admissible aggregator* if it is *grounded*, meaning $A(0) = 0$; *monotone*, meaning $u \leq v$ coordinatewise implies $A(u) \leq A(v)$; and *subadditive*, meaning $A(u+v) \leq A(u) + A(v)$ for all $u, v \in \mathbb{R}_{\geq 0}^n$. It is *strict* if in addition $A(u) > 0$ whenever $u \neq 0$.

Theorem 5.2 (General aggregation theorem). *Let $\delta_1, \dots, \delta_n$ be pseudometrics on $\mathcal{X}$ and let A be an admissible aggregator. Then $d_A(x, y) := A(\delta_1(x, y), \dots, \delta_n(x, y))$ is a pseudo-metric on $\mathcal{X}$; it is a metric if A is strict and the views separate points; and if some δ_i are quasi-metrics then d_A is a quasi-pseudo-metric.*

Proof. Grounding gives $d_A(x, x) = A(0) = 0$ and the codomain gives $d_A \geq 0$, while symmetry is inherited coordinatewise. For the triangle inequality write $a_i = \delta_i(x, y)$, $b_i = \delta_i(y, z)$ and $c_i = \delta_i(x, z)$, so that $c \leq a + b$ coordinatewise; then monotonicity and subadditivity give $d_A(x, z) = A(c) \leq A(a + b) \leq A(a) + A(b) = d_A(x, y) + d_A(y, z)$. If A is strict and the views separate points then $x \neq y$ forces some $\delta_i(x, y) > 0$ and hence $d_A(x, y) > 0$. No step used symmetry, which gives the quasi-metric case. $\square$

Theorem 4.3 is the special case in which

$$A(u) = \left(\sum_i w_i u_i^p \right)^{1/p},$$

whose subadditivity is precisely Minkowski’s inequality. Isolating the three conditions is worthwhile. It turns the question of which aggregation rules are permitted from a case-by-case investigation into a three-line check, and admissibility is preserved by exactly the operations one wants to perform.

Proposition 5.3 (Closure properties). *Let A, B be admissible aggregators and $\lambda, \mu \geq 0$. Then $\lambda A + \mu B$ is admissible; $\max(A, B)$ is admissible; $\psi \circ A$ is admissible for any non-decreasing concave ψ with $\psi(0) = 0$; and if $A_1, \dots, A_m$ are admissible on disjoint groups of coordinates and A_0 is admissible on $\mathbb{R}_{\geq 0}^m$ then the nested map $u \mapsto A_0(A_1(u^{(1)}), \dots, A_m(u^{(m)}))$ is admissible.*

Proof. Each of the three defining properties is preserved by non-negative linear combination. For the maximum, $\max(A, B)(u + v) \leq \max(A(u) + A(v), B(u) + B(v)) \leq \max(A, B)(u) + \max(A, B)(v)$ using non-negativity, and grounding and monotonicity are clear. For the concave reshaping, grounding and monotonicity compose, and $\psi(A(u + v)) \leq \psi(A(u) + A(v)) \leq \psi(A(u)) + \psi(A(v))$ by monotonicity and the subadditivity of concave ψ established in Lemma 3.3. For nesting, apply the inner aggregators’ subadditivity, then the monotonicity of A_0 , then the subadditivity of A_0 . $\square$

Nesting is the practically important closure. It licenses group structure in the geometry: aggregate the semantic sub-views with $p = 1$ because they are substitutable evidence, aggregate the policy views with $p = \infty$ because they are hard constraints, then aggregate the groups with $p = 2$. The result remains a metric and Part II applies verbatim. This is strictly more expressive than a flat L^p over all views, and it matches how practitioners actually reason about relevance — as a small hierarchy of considerations rather than a flat list.

Example 5.4 (Admissible aggregators in use). The weighted power mean with $p \geq 1$ is the default of Definition 4.1. The weighted Chebyshev rule $A(u) = \max_i w_i u_i$ is non-compensatory while retaining weights, unlike the limit $p \rightarrow \infty$, which discards weight magnitudes as Remark 5.6 explains. The ordered weighted average $A(u) = \sum_j \omega_j u_{(j)}$ with u sorted decreasingly and ω non-increasing expresses “the worst few views are what matter”. The Choquet integral with respect to a submodular capacity encodes redundancy between views, so that two nearly duplicate semantic views do not count twice. Finally the smooth surrogate $A(u) = \beta^{-1} \log \sum_i e^{\beta w_i u_i}$, grounded by subtracting its value at zero, approximates the Chebyshev rule differentiably.

Proposition 5.5 (Ordered weighted averages and Choquet integrals). *Let $\omega_1 \geq \dots \geq \omega_n \geq 0$ and let $A(u) = \sum_j \omega_j u_{(j)}$ be the corresponding ordered weighted average. Then A is admissible. More generally the Choquet integral with respect to a submodular capacity ν with $\nu(\emptyset) = 0$ is admissible.*

Proof. Grounding and monotonicity are clear from $\omega \geq 0$. For subadditivity the rearrangement inequality gives $A(u) = \max_{\pi} \sum_j \omega_j u_{\pi(j)}$, a pointwise maximum of linear functionals, hence convex and positively homogeneous, hence sublinear: $A(u + v) = 2A(\frac{u+v}{2}) \leq 2(\frac{1}{2}A(u) + \frac{1}{2}A(v)) = A(u) + A(v)$. The Choquet integral with respect to a submodular capacity is the Lovász extension of ν , which is convex and positively homogeneous, so the same argument applies. $\square$

Remark 5.6 (Weighted Chebyshev versus the limit in p). By Theorem 4.6 the weighted power mean converges to $\max_{i \in \text{supp}(w)} u_i$, so in the limit the weights select which views participate but no longer scale them. If graded importance is wanted at the bottleneck,

use $A(u) = \max_i w_i u_i$ directly; it is obtained by folding the weights into the views and is admissible by Proposition 5.3.

5.2 In what sense L^p is canonical

Proposition 5.7 (Canonicity of the power means). *Let $A(u) = f^{-1}(\sum_i w_i f(u_i))$ be a weighted quasi-arithmetic mean with f continuous, strictly increasing and $f(0) = 0$, and w a probability vector. If A is positively homogeneous then $f(t) = ct^p$ for some $p > 0$, so that A is a weighted power mean [16]; and among these, A is subadditive, hence admissible, if and only if $p \geq 1$.*

Proof. The first statement is the classical characterisation of homogeneous quasi-arithmetic means. Sufficiency in the second is Lemma 4.2. For necessity let $p < 1$ and take $n = 2$, $w = (\frac{1}{2}, \frac{1}{2})$, $u = (1, 0)$ and $v = (0, 1)$; then $M_p(u) = M_p(v) = (\frac{1}{2})^{1/p}$ while $M_p(u + v) = M_p(1, 1) = 1$, so subadditivity would require $1 \leq 2^{1-1/p}$, that is $p \geq 1$. At $p = \frac{1}{2}$ the numbers are 1 against $\frac{1}{4} + \frac{1}{4}$. $\square$

Within the class of scale-invariant means, therefore, the restriction $p \geq 1$ is a requirement rather than a preference, and the admissible sub-family is exactly the one parameterised in Definition 16.1. Departures from L^p — ordered weighted averages, Choquet integrals, nesting — are departures from quasi-arithmetic averaging rather than from homogeneity, and Proposition 5.3 covers them.

5.3 Ultrametric views and canonical context partitions

One family deserves separate treatment because it changes the qualitative behaviour of neighbourhoods rather than merely reweighting them.

Definition 5.8 (Ultrametric view). A view δ is an *ultrametric* if it satisfies the strong triangle inequality $\delta(x, z) \leq \max(\delta(x, y), \delta(y, z))$.

Ultrametrics arise wherever objects carry a tree or a lineage: directory paths, module hierarchies, commit ancestry, semantic version numbers, taxonomy codes, organisational charts, and the dendrogram of any hierarchical clustering. The canonical arithmetic example is the p -adic absolute value $|x - y|_p = p^{-v_p(x-y)}$ with v_p the p -adic valuation, so that encoding a hierarchical identifier as a p -adic integer makes the distance a strictly decreasing function of shared-prefix depth and “same major version” dominates every lower-order difference without weight tuning [19].

Theorem 5.9 (Ultrametric aggregation and canonical partition). *Let $\delta_1, \dots, \delta_n$ be ultrametrics and let $d(x, y) = \max_i w_i \delta_i(x, y)$ with $w_i \geq 0$. Then d is an ultrametric and in particular a pseudometric; every point of a ball is a centre, so that $d(x, y) < \varepsilon$ implies $B_\varepsilon(y) = B_\varepsilon(x)$; the balls of radius ε therefore partition $\mathcal{X}$ into disjoint blocks and any two balls are nested or disjoint; and consequently, if the agent’s state moves from s to s' with $d(s, s') < \varepsilon$, then $N_\varepsilon(s') = N_\varepsilon(s)$ exactly, so the context need not be recomputed at all.*

Proof. Scaling by $w_i \geq 0$ preserves the strong triangle inequality, and for the maximum, $w_i \delta_i(x, z) \leq \max(w_i \delta_i(x, y), w_i \delta_i(y, z)) \leq \max(d(x, y), d(y, z))$ for each i , so taking the maximum over i gives $d(x, z) \leq \max(d(x, y), d(y, z))$, which implies the ordinary triangle inequality because $\max(a, b) \leq a + b$ for $a, b \geq 0$. If $d(x, y) < \varepsilon$ and $z \in B_\varepsilon(y)$ then $d(x, z) \leq \max(d(x, y), d(y, z)) < \varepsilon$, so $B_\varepsilon(y) \subseteq B_\varepsilon(x)$, and exchanging x and y gives equality. If two balls of a common radius intersect, pick z in the intersection; both then equal $B_\varepsilon(z)$, so distinct balls are disjoint and cover $\mathcal{X}$, while nesting for distinct radii

follows from monotonicity of balls in ε . The last claim is the second with $x = s$ and $y = s'$. $\square$

Exact reuse rather than bounded drift. The final clause is strictly stronger than the generic stability bound of Theorem 6.7. In the metric case a small state change perturbs the context only within an annulus; in the ultrametric case it leaves the context identical until the state crosses a block boundary. For agents that spend many consecutive steps inside one module, one package version or one work item, an ultrametric structural view therefore converts context construction from a per-step computation into a per-block one. Section 19 measures 86.3% exact reuse on a simulated hierarchical workspace against 18.1% for a flat geometry of the same nominal radius, which is the single largest efficiency effect we observed.

5.4 Component distances by object type

Which distance belongs on which view is an empirical question, but the space of sensible answers is well charted. Table 3 summarises the recommendation and Appendix B gives the full catalogue with formulas, metric class, ranges and intended use.

Object / view	Recommended distance	Why
Text, embeddings	angular distance; $\sqrt{2 - 2\cos}$	genuine metrics, unlike $1 - \cos$
Distributions, outcome profiles	Hellinger; Jensen–Shannon distance; Wasserstein-1	metrics; JS bounded, W_1 geometry-aware
Sets, tags, permissions	Jaccard; Hamming	metrics; Jaccard bounded in $[0, 1]$
Code, ASTs, configs	tree edit distance; token Jaccard; normalised compression distance	structural rather than lexical
Dependency / call graphs	shortest path; resistance; diffusion distance	resistance and diffusion tolerate single-edge noise
Hierarchies, versions, lineage	ultrametric from the tree; p -adic; hyperbolic	Theorem 5.9; low-distortion tree embedding
Time, recency	$\psi(\Delta t)$; circular distance for periodicity	bounded, concave, metric-preserving
Rankings, plans	Kendall tau; Spearman footrule	metrics on permutations; stability measurement
Whole environments	Gromov–Hausdorff; Gromov–Wasserstein	compares spaces whose object sets differ

Table 3: Recommended component distance by view type; full catalogue in Appendix B.

5.5 Admitting non-metric dissimilarities

Several useful comparisons are not metrics: Kullback–Leibler and other Bregman divergences, dynamic time warping, Bray–Curtis, Jaro–Winkler, SimRank, and any raw “one minus similarity”. TopoSpace admits them, but never silently, and four metrisation recipes are available in decreasing order of preference.

Square root of a squared-Hilbertian divergence. If δ^2 is squared-Hilbertian, as for Jensen–Shannon, χ^2 , Hellinger and the energy distance, then δ itself is a metric; the Jensen–Shannon case is the standard example [14].

Kernel metrification. If a positive semidefinite kernel κ is available, use $\sqrt{2 - 2\kappa}$ by Lemma 3.4. This covers Weisfeiler–Lehman graph kernels [30] and MMD-based comparisons [15].

Concave reparameterisation. Apply $\psi(t) = t/(1+t)$ or $\psi(t) = 1 - e^{-\lambda t}$ to a valid metric to bound it, by Lemma 3.3. This normalises a metric; it does not repair a non-metric, and confusing the two is a common error.

Symmetrisation and chaining. For a quasi-metric q use $\frac{1}{2}(q(x,y) + q(y,x))$, which is a pseudometric, when symmetry is needed; or keep the asymmetry and work with the chain distance of Proposition 6.4, which is the right choice when the directionality is itself informative.

When no recipe applies the view is declared raw and the runtime downgrades the guarantees automatically. Table 4 records what is lost, which is the information a practitioner needs in order to decide whether the downgrade is acceptable.

Result	Metric	Pseudo	Quasi	Raw
Neighbourhood construction, ranking (Def. 7.1)	✓	✓	✓	✓
Exact recovery under a margin (Thm. 7.3)	✓	✓	✓	✓
Neighbourhood stability (Thm. 6.7)	✓	✓	✓	✓
Constraint enforcement (Thm. 11.1)	✓	✓	✓	✓
Sublevel-set optimality (Thm. 7.4)	✓	✓	✓	✓
Deduplication by metric quotient (Cor. 4.4)	✓	✓	—	—
Chain distance and global consistency (Prop. 6.4)	✓	✓	✓	—
Delegation containment (Prop. 12.1)	✓	✓	—	—
Filtration, persistence, stability (Thm. 8.4)	✓	✓	—	—
Topological drift as a pseudometric (Thm. 15.3)	✓	✓	—	—
Index filtering bound (Thm. 17.2)	✓ [†]	✓ [†]	—	—
Canonical partition, exact reuse (Thm. 5.9)	ultrametric views only			

Table 4: Which guarantees survive which class of component dissimilarity. [†] additionally requires the coordinate form assumed in Theorem 17.2.

The table is the operational payoff of this section. Ranking, radius selection, exact recovery and constraint enforcement are robust to almost any dissimilarity a practitioner might plug in, so experimentation is cheap; the compositional results — delegation, planning consistency, persistence, drift — are exactly the ones a raw similarity silently destroys. A runtime that tracks the class of each admitted view can therefore report which theorems still apply to the current configuration, and the reference implementation does so.

5.6 Cross-space distances and a cleaner definition of drift

Definition 15.2 compares environments by embedding both object sets into their union and assigning a large distance to missing objects. That convention is serviceable but arbitrary; the intrinsic alternative compares the environments as metric measure spaces.

Proposition 5.10 (Intrinsic environment distance). *Let $\mathcal{A} = (\mathcal{X}, d, \mu)$ and $\mathcal{A}' = (\mathcal{X}', d', \mu')$ be finite metric measure spaces whose measures encode object importance normalised to probability. The Gromov–Wasserstein distance*

$$\text{GW}_q(\mathcal{A}, \mathcal{A}') = \left(\inf_{\pi \in \Pi(\mu, \mu')} \int \int |d(x, y) - d'(x', y')|^q d\pi(x, x') d\pi(y, y') \right)^{1/q}$$

is a metric on isomorphism classes of such spaces [24], is defined without any common index set, and reduces to the uniform metric drift up to constants when $\mathcal{X} = \mathcal{X}'$ and π is the identity coupling. Using it as the drift statistic removes the embedding convention while preserving Theorem 15.3 and Proposition 15.6, both of which require only the metric axioms.

Proof. Metricity on isomorphism classes is [24]. For the reduction, take $\mathcal{X} = \mathcal{X}'$ and the identity coupling, admissible when $\mu = \mu'$; the objective becomes the $L^q(\mu \otimes \mu)$ norm of $d - d'$, bounded by $\sup_{x, y} |d(x, y) - d'(x, y)|$. The cited results transfer because their proofs use only symmetry, the triangle inequality and non-negativity. $\square$

The cost is computational: Gromov–Wasserstein requires solving a quadratic assignment relaxation, typically by entropic regularisation [27], and is therefore reserved for periodic environment audits rather than per-step monitoring, for which the Jaccard statistic of Section 15 remains the default.

6 Adaptive Geometry

6.1 Parameterisation

Definition 6.1 (Adaptive geometry). Let Θ be a parameter set. An *adaptive geometry* is a triple of measurable maps $p_\theta : \mathcal{S} \times \mathcal{G} \rightarrow [1, \infty]$, $w_\theta : \mathcal{S} \times \mathcal{G} \rightarrow \Delta^{n-1}$ and $\varepsilon_\theta : \mathcal{S} \times \mathcal{G} \times 2^{\mathcal{X}} \rightarrow (0, \infty]$, where Δ^{n-1} is the probability simplex. Given (s, g) we write $p_s := p_\theta(s, g)$ and $w_s := w_\theta(s, g)$ and define the *state-anchored dissimilarity*

$$D_\theta(s, x) := d_{p_s, w_s}(s, x), \quad x \in \mathcal{X}. \quad (4)$$

Proposition 6.2 (Per-step metricity, global asymmetry). *For each fixed (s, g) the function d_{p_s, w_s} is a pseudometric on $\mathcal{X} \cup \{s\}$ by Theorem 4.3. The two-argument function $(x, y) \mapsto D_\theta(x, y)$ obtained by letting the first argument determine the geometry is in general not symmetric: there exist view systems and parameter maps with $D_\theta(s, x) \neq D_\theta(x, s)$.*

Proof. The first claim is Theorem 4.3 with the fixed parameters (p_s, w_s) . For the second take $n = 2$ with $\delta(s, x) = (1, 0)$ and parameter maps giving $w_s = (1, 0)$ and $w_x = (0, 1)$; then $D_\theta(s, x) = 1$ while $D_\theta(x, s) = 0$. $\square$

Asymmetry is a feature, but it must be paid for. Proposition 6.2 is not a defect to be repaired. Relevance is directional: the stack trace is relevant to the debugging agent, while the debugging agent is not relevant to the stack trace in the same sense, and systems that force symmetry onto relevance are making a modelling error. The price of

admitting asymmetry is that we must state which results require symmetry. The filtration and persistence results of Section 8 do require it; the neighbourhood, recovery, routing and constraint results do not; and Table 4 records the split.

6.2 Metric fields and a globally consistent distance

Results that do require a single global object — notably planning, where distances must compose along a trajectory — can always be supplied with one.

Definition 6.3 (Metric field and chain distance). A *metric field* on $\mathcal{X}$ is an assignment $z \mapsto d_z$ of a pseudometric to each point. Its *chain distance* is

$$\rho(x, y) = \inf \left\{ \sum_{j=0}^{m-1} d_{z_j}(z_j, z_{j+1}) : m \in \mathbb{N}, z_0 = x, z_m = y, z_j \in \mathcal{X} \right\}.$$

Proposition 6.4 (Chain distance). *The chain distance satisfies $\rho(x, x) = 0$, $\rho \geq 0$ and $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$; it obeys $\rho(x, y) \leq d_x(x, y)$ and is the largest function with these properties. If the field is uniformly comparable, meaning there are $0 < \alpha \leq \beta < \infty$ and a reference pseudometric $d_\star$ with $\alpha d_\star \leq d_z \leq \beta d_\star$ for all z , then $\alpha d_\star \leq \rho \leq \beta d_\star$, so ρ is bi-Lipschitz equivalent to $d_\star$ and induces the same topology.*

Proof. The trivial chain gives $\rho(x, x) = 0$; concatenating an admissible chain from x to y with one from y to z gives an admissible chain from x to z of the summed length, and taking infima yields the triangle inequality. The single-step chain gives $\rho(x, y) \leq d_x(x, y)$. If ρ' satisfies the triangle inequality and is dominated stepwise then along any chain $\rho'(x, y) \leq \sum_j d_{z_j}(z_j, z_{j+1})$, so $\rho' \leq \rho$. Under uniform comparability every chain has length at least $\alpha \sum_j d_\star(z_j, z_{j+1}) \geq \alpha d_\star(x, y)$ by the triangle inequality for $d_\star$, and the upper bound comes from the single-step chain. $\square$

Proposition 6.4 supplies the object needed in Section 13. Even when the geometry is state-dependent there is a canonical globally consistent distance beneath it, and under uniform comparability — which the parameterisation of Section 16 enforces by keeping the weights away from the boundary of the simplex — it is equivalent to any fixed reference metric.

6.3 Stability under parameter perturbation

Learning changes θ , and caching assumes that small changes in θ do not invalidate everything. The next three results make this precise, and Section 19 reports that both bounds held on every one of three hundred randomised trials.

Theorem 6.5 (Weight stability). *Let $p \in [1, \infty)$, let $w, w' \in \mathbb{R}_{\geq 0}^n$ and let $a \in \mathbb{R}_{\geq 0}^n$ be a view vector with $\|a\|_\infty \leq 1$. Then*

$$|d_{p,w} - d_{p,w'}| \leq \|a\|_\infty \|w - w'\|_1^{1/p} \leq \|w - w'\|_1^{1/p}.$$

Proof. Write $u_i = w_i^{1/p} a_i$ and $v_i = w_i'^{1/p} a_i$, so that $d_{p,w} = \|u\|_p$ and $d_{p,w'} = \|v\|_p$ in the unweighted ℓ_p norm. The reverse triangle inequality gives $|\|u\|_p - \|v\|_p| \leq \|u - v\|_p$ and

$$\|u - v\|_p = \left(\sum_i |w_i^{1/p} - w_i'^{1/p}|^p a_i^p \right)^{1/p} \leq \|a\|_\infty \left(\sum_i |w_i^{1/p} - w_i'^{1/p}|^p \right)^{1/p}.$$

For $p \geq 1$ and $\alpha, \beta \geq 0$ we have $|\alpha^{1/p} - \beta^{1/p}| \leq |\alpha - \beta|^{1/p}$, since $t \mapsto t^{1/p}$ is concave on $\mathbb{R}_{\geq 0}$ with value 0 at 0 and hence subadditive; therefore $\sum_i |w_i^{1/p} - w_i'^{1/p}|^p \leq \|w - w'\|_1$. $\square$

Proposition 6.6 (Exponent stability and the exact derivative). *With w a probability vector and $a \in [0, 1]^n$ we have $|M_p(a) - M_q(a)| \leq M_\infty(a) - M_1(a)$ for all $p, q \in [1, \infty]$. Moreover, wherever $S = \sum_i w_i a_i^p > 0$ the derivative exists and equals*

$$\frac{\partial M_p}{\partial p} = M_p \left[\frac{1}{p} \cdot \frac{\sum_i w_i a_i^p \ln a_i}{S} - \frac{\ln S}{p^2} \right], \quad (5)$$

with the convention $a_i^p \ln a_i = 0$ when $a_i = 0$.

Proof. The bound is Theorem 4.6. For the derivative write $\ln M_p = \frac{1}{p} \ln S(p)$ with $S(p) = \sum_i w_i a_i^p$ and $S'(p) = \sum_i w_i a_i^p \ln a_i$; differentiating gives $\partial_p \ln M_p = -\frac{1}{p^2} \ln S + \frac{1}{p} \frac{S'}{S}$, and multiplying by M_p yields (5). The convention at $a_i = 0$ is justified by $\lim_{t \downarrow 0} t^p \ln t = 0$ for $p > 0$. $\square$

Theorem 6.7 (Neighbourhood stability). *Let D, D' map $\mathcal{X}$ into $\mathbb{R}_{\geq 0}$ and let $\varepsilon, \varepsilon' > 0$, with*

$$\sup_x |D(x) - D'(x)| \leq \eta, \quad |\varepsilon - \varepsilon'| \leq \eta'.$$

Put

$$\mathbf{N} = \{x : D(x) < \varepsilon\}, \quad \mathbf{N}' = \{x : D'(x) < \varepsilon'\}, \quad \tau = \eta + \eta'.$$

Then

$$\{x : D(x) < \varepsilon - \tau\} \subseteq \mathbf{N}' \subseteq \{x : D(x) < \varepsilon + \tau\},$$

and therefore the symmetric difference is confined to an annulus,

$$\mathbf{N} \triangle \mathbf{N}' \subseteq A_\tau := \{x : \varepsilon - \tau \leq D(x) \leq \varepsilon + \tau\}.$$

In particular, if no object lies in A_τ then $\mathbf{N}' = \mathbf{N}$ exactly.

Proof. If $D(x) < \varepsilon - \tau$ then $D'(x) \leq D(x) + \eta < \varepsilon - \eta' \leq \varepsilon'$, so $x \in \mathbf{N}'$; and if $x \in \mathbf{N}'$ then $D(x) \leq D'(x) + \eta < \varepsilon' + \eta \leq \varepsilon + \tau$. For the symmetric difference, any $x \in \mathbf{N} \setminus \mathbf{N}'$ has $D(x) < \varepsilon$ and, by the contrapositive of the first inclusion, $D(x) \geq \varepsilon - \tau$; any $x \in \mathbf{N}' \setminus \mathbf{N}$ has $D(x) \geq \varepsilon$ and, by the second inclusion, $D(x) < \varepsilon + \tau$. If $A_\tau \cap \mathcal{X} = \emptyset$ the symmetric difference is empty. $\square$

Three engineering practices become theorems. Theorem 6.7 is the formal justification for behaviour that would otherwise be folklore. Context caching across consecutive steps is safe whenever the state moves by less than the annulus width. Online learning of θ does not cause erratic context churn, because the context can only change for objects near the boundary. And placing ε in a gap of the distance spectrum, as the radius rules of Section 7 do, makes the selected set robust to parameter noise of magnitude up to half the gap. Our stability experiment measured a mean symmetric-difference fraction of 0.113 under a weight perturbation of $\|w - w'\|_1 = 0.05$, with every observed difference contained in the predicted annulus on all three hundred trials.

7 Adaptive Neighbourhoods and the Choice of Radius

7.1 Neighbourhoods

Definition 7.1 (Adaptive neighbourhood). For a state s , goal g and object set $\mathcal{X}$,

$$\mathbf{N}_\theta(s, g) = \{x \in \mathcal{X} : D_\theta(s, x) < \varepsilon_\theta(s, g, \mathcal{X})\},$$

called the *context* of the agent at that step; its cardinality is the *width*.

The design turns on the fact that the width is an output. We consider five radius rules, all implemented, with the choice among them itself part of θ . Let $D_{(1)} \leq \dots \leq D_{(N)}$ be the sorted anchored distances.

R1, the rank radius. $\varepsilon = D_{(k)} + 0^+$ for a fixed k ; this is Top- k in disguise and is included as the baseline inside the family.

R2, the quantile radius. ε is the q -quantile of the spectrum for a fixed q ; it adapts to the distance distribution but not to its shape.

R3, the gap radius. $\varepsilon = \frac{1}{2}(D_{(j^*)} + D_{(j^*+1)})$ where $j^* = \arg \max_{j \leq j_{\max}} (D_{(j+1)} - D_{(j)})$ locates the largest gap among the leading candidates. This is the rule that exploits Theorem 7.3 and the one that performed best in our study.

R4, the self-tuning radius. $\varepsilon = c D_{(k)}$ for a fixed $c > 1$ and small k , in the spirit of self-tuning spectral clustering [36]; it is scale invariant and cheap but insensitive to the gap structure.

R5, the budget radius. ε is the largest value whose sublevel set fits a token or latency budget, which makes the rule directly answerable to the operational constraint rather than to a statistical one.

Proposition 7.2 (Top- k is the degenerate rule). *Under R1 with $n = 1$ and δ_1 the angular distance, $\mathbf{N}_\theta(s, g)$ equals the embedding Top- k set of (2) with ties broken identically. The family of adaptive neighbourhoods therefore contains Top- k retrieval, and the containment is strict because R3 produces a width that varies with s while R1 does not.*

Proof. With $n = 1$ the aggregate is $d_{p,w}(s, x) = w_1^{1/p} \delta_1(s, x)$ for every p , a strictly increasing function of δ_1 , so the induced order is the order by angular distance, which for unit-normalised embeddings reverses the cosine order; selecting the objects with $D < D_{(k)} + 0^+$ returns exactly the k smallest. Strictness holds because under R3 the width equals j^* , which depends on the realised spectrum and hence on s . $\square$

7.2 Exact recovery under a margin

Theorem 7.3 (Exact recovery). *Suppose there is a ground-truth relevant set $R \subseteq \mathcal{X}$ and a margin $\alpha < \beta$ with $D_\theta(s, x) \leq \alpha$ for all $x \in R$ and $D_\theta(s, x) \geq \beta$ for all $x \in \mathcal{X} \setminus R$. Then for every $\varepsilon \in (\alpha, \beta]$ we have $\mathbf{N}_\theta(s, g) = R$ exactly, so precision and recall are both 1. Rule R3 attains this whenever the largest gap in the spectrum is the one separating R from its complement, in particular whenever $\beta - \alpha$ exceeds every gap internal to either block. Any fixed- k rule attains it only if $k = |R|$, and if $k \neq |R|$ its F_1 score is at most $2 \min(k, |R|) / (k + |R|) < 1$. Finally the conclusion is robust: if D_θ is perturbed by at most $\eta < (\beta - \alpha)/2$ and ε is placed at the midpoint, then still $\mathbf{N}_\theta(s, g) = R$.*

Proof. For $\varepsilon \in (\alpha, \beta]$ every $x \in R$ has $D(x) \leq \alpha < \varepsilon$ and is selected while every $x \notin R$ has $D(x) \geq \beta \geq \varepsilon$ and is not. For the gap rule, the margin places all of R before all of its complement in the sorted order with a boundary gap of at least $\beta - \alpha$, so if that exceeds every internal gap then $j^* = |R|$ and ε falls strictly between the blocks. A fixed- k rule returns the k nearest objects, which by the margin are the first $\min(k, |R|)$ elements of R followed by elements of the complement, so $|\mathbf{N} \cap R| = \min(k, |R|)$, precision is $\min(k, |R|)/k$, recall is $\min(k, |R|)/|R|$ and their harmonic mean is $2 \min(k, |R|)/(k + |R|)$, equal to 1 only when $k = |R|$. For robustness, if $|\tilde{D} - D| \leq \eta$ then $x \in R$ gives $\tilde{D}(x) \leq \alpha + \eta < (\alpha + \beta)/2$ and $x \notin R$ gives $\tilde{D}(x) \geq \beta - \eta > (\alpha + \beta)/2$, so the midpoint threshold separates them; this is Theorem 6.7 with an empty annulus. $\square$

What the theorem buys, and what it assumes. Under a margin the adaptive rule is exactly correct and the fixed- k rule is provably not, for every k except the one a human could have chosen only by knowing the answer in advance. The margin hypothesis is of course an assumption about the data, but it is precisely the assumption that a well-chosen geometry is supposed to create, which is why learning θ and choosing ε are two halves of one problem rather than two independent tuning exercises. Corollary 16.5 closes the loop by showing that a zero hinge loss on logged data certifies the margin, and Section 19 measures the resulting exact-recovery rate at 16.7% for R3 against 7.1%, 3.2% and 2.7% for the best fixed choices $k = 3$, $k = 5$ and $k = 10$.

7.3 Decision-theoretic optimality of ball-shaped context

Theorem 7.4 (Sublevel-set optimality). *Suppose each object x carries an expected usefulness $u(x) \in \mathbb{R}$ and a marginal cost $\lambda > 0$ per object included, and that usefulness is a non-increasing function of distance, $u(x) = h(D_\theta(s, x))$ with h non-increasing. Then the subset maximising the net utility $U(S) = \sum_{x \in S} u(x) - \lambda |S|$ over all $S \subseteq \mathcal{X}$ is the sublevel set $S^* = \{x : h(D_\theta(s, x)) > \lambda\}$, which equals $\{x : D_\theta(s, x) < \varepsilon\}$ for $\varepsilon = \sup\{t : h(t) > \lambda\}$.*

Proof. $U(S) = \sum_{x \in S} (u(x) - \lambda)$ decomposes over objects, so it is maximised by including exactly those x with $u(x) - \lambda > 0$, ties being immaterial. Since h is non-increasing, that set is a sublevel set of D with the stated threshold. $\square$

Theorem 7.4 answers a question the ball construction invites: why should the relevant region be a ball rather than some other shape? Under a monotone usefulness–distance relationship the optimal shape *is* a ball, whose radius sits where marginal usefulness equals marginal cost. If the monotone relationship fails then the geometry is wrong and should be re-learned; the correct response is not to abandon balls but to change the metric that defines them.

7.4 Invariance

Proposition 7.5 (Scale invariance). *Rules R1, R3 and R4 are invariant under a uniform rescaling $D \mapsto cD$ with $c > 0$; rules R2 and R5 are not in general.*

Proof. R1 is defined by rank. Under $D \mapsto cD$ the sorted spectrum becomes $(cD_{(j)})_j$, so R4's threshold transforms as $c'D_{(k)} \mapsto cc'D_{(k)}$ and R3's largest-gap index is unchanged with the midpoint scaling by c ; in both cases $\{x : cD(x) < c\varepsilon\} = \{x : D(x) < \varepsilon\}$. R2 depends on the quantile of the value distribution and R5 on an absolute budget, neither of which need be preserved. $\square$

Scale invariance matters because the absolute scale of D_θ has no meaning: it changes whenever weights are renormalised or a view is recalibrated. A radius rule that is not scale invariant must be recalibrated after every such change, which in practice means it will silently drift out of calibration, and this is a strong argument for R3 and R4 as defaults.

8 Multi-Scale Context and Filtrations

8.1 Nested contexts

A single radius forces a binary decision on objects whose usefulness is graded. Agents in practice want a tiered context: a small core of immediately actionable material, a wider band of supporting evidence, and a still wider band of background that should be summarised rather than quoted.

Definition 8.1 (Neighbourhood filtration). For $0 < \varepsilon_1 < \dots < \varepsilon_m$ the *neighbourhood filtration* at s is the nested family $\mathbf{N}^{(\ell)}(s) = \{x : D_\theta(s, x) < \varepsilon_\ell\}$ with $\mathbf{N}^{(1)} \subseteq \dots \subseteq \mathbf{N}^{(m)}$.

Proposition 8.2 (Monotonicity and entry scale). *The inclusions hold and each object has an entry scale $\beta(x) = D_\theta(s, x)$ with $x \in \mathbf{N}^{(\ell)}$ if and only if $\varepsilon_\ell > \beta(x)$. The filtration is therefore described completely by the multiset $\{\beta(x)\}_{x \in \mathcal{X}}$, and no object leaves once it has entered.*

Proof. Immediate from $\varepsilon_\ell < \varepsilon_{\ell+1}$ and the definition of a sublevel set. $\square$

Proposition 8.2 implies that the filtration of *points* alone has trivial persistence, since every object is born and never dies. Non-trivial topology requires a complex rather than a set, which is the subject of the next subsection and the only place in the framework where homology does genuine work.

8.2 The Vietoris–Rips complex of a context

Definition 8.3 (Context complex). Assume the views are symmetric so that $d_{p,w}$ is a pseudometric on $\mathcal{X}$. For $\varepsilon > 0$ let $\text{VR}_\varepsilon(s)$ be the Vietoris–Rips complex on the vertex set $\mathbf{N}^{(\varepsilon)}(s)$ whose k -simplices are the $(k+1)$ -subsets of pairwise distance below ε ; its persistent homology over the scale parameter yields the persistence diagram $\text{Dgm}(s)$ [11, 3].

The homology has a direct reading in agent terms. Connected components of the complex are topic clusters within the context, so a context that splits into three long-lived components is one in which the agent is being handed three separate stories — useful both for summarisation, which should then proceed per component, and for diagnosis, since a context that ought to be coherent and is not usually indicates a mis-specified goal. One-dimensional cycles indicate contexts in which evidence forms a loop without a filling, which is the signature of competing explanations that are pairwise close but admit no common refinement.

Theorem 8.4 (Stability of the context diagram). *Let d, d' be pseudometrics on a finite set V with $\|d - d'\|_\infty \leq \eta$. Then the Vietoris–Rips filtrations are η -interleaved and $d_B(\text{Dgm}(d), \text{Dgm}(d')) \leq \eta$.*

Proof. If $|d - d'| \leq \eta$ pointwise then every simplex of VR_ε^d lies in $\text{VR}_{\varepsilon+\eta}^{d'}$ and conversely, giving inclusions $\text{VR}_\varepsilon^d \subseteq \text{VR}_{\varepsilon+\eta}^{d'} \subseteq \text{VR}_{\varepsilon+2\eta}^d$ that commute with the filtration maps. This is an η -interleaving of the persistence modules, and algebraic stability [4, 6] converts it into the bottleneck bound. $\square$

Corollary 8.5 (Parameter-perturbation stability). *If the weights change by $\|w - w'\|_1 \leq \zeta$ with p fixed and all view values bounded by 1, then $d_B(\text{Dgm}(w), \text{Dgm}(w')) \leq \zeta^{1/p}$ by Theorems 6.5 and 8.4. If instead the exponent changes with w fixed then $d_B \leq \sup_{x,y} (M_\infty(x, y) - M_1(x, y))$.*

These results are what make the drift theory meaningful. Without stability, a change in the diagram could not be attributed to a change in the environment, since it might equally reflect a negligible change in parameters; with it, a detector can be given a threshold that provably cannot fire on parameter updates alone, as Proposition 15.5 does.

8.3 Tiered context assembly

Definition 8.6 (Tiered assembly). Given the filtration, the runtime assembles the prompt from $\mathbf{N}^{(1)}$ verbatim, from $\mathbf{N}^{(2)} \setminus \mathbf{N}^{(1)}$ in compressed form, and from $\mathbf{N}^{(3)} \setminus \mathbf{N}^{(2)}$ as an index only, subject to a token budget B split as $B_1 \geq B_2 \geq B_3$ with $B_1 + B_2 + B_3 \leq B$.

Proposition 8.7 (Budget feasibility). *Let the per-object token cost at tier ℓ be non-increasing in ℓ . The assembly is feasible for some choice of radii whenever the cheapest tier can hold the whole universe, and the map from ε_1 to total cost is non-decreasing and piecewise constant with at most N pieces, so the largest feasible ε_1 is found by binary search in $O(N \log N)$ time after sorting.*

Proof. Total cost sums a tier-dependent per-object cost; increasing ε_1 moves objects to strictly more expensive tiers, so cost is non-decreasing, and it changes only when ε_1 crosses an entry scale, giving piecewise constancy. Monotone piecewise-constant functions admit binary search. $\square$

9 An Axiomatic Characterisation of Context Selection

We have proposed that context be a ball. It is fair to ask whether that is a substantive commitment or one heuristic among many. Under three natural axioms it is a theorem.

Definition 9.1 (Selection rule). A *selection rule* assigns to each state s , with its goal suppressed from the notation, a subset $\sigma(s) \subseteq \mathcal{X}$.

Axiom 9.2 (Locality). There is a function $D : \mathcal{S} \times \mathcal{X} \rightarrow \mathbb{R}$ such that $\sigma(s)$ depends on $\mathcal{X}$ only through the values $\{D(s, x)\}_{x \in \mathcal{X}}$.

Axiom 9.3 (Anonymity). σ commutes with relabelling: for any bijection of $\mathcal{X}$ preserving $D(s, \cdot)$, the set $\sigma(s)$ is invariant.

Axiom 9.4 (Threshold consistency). If $x \in \sigma(s)$ and $D(s, y) \leq D(s, x)$ then $y \in \sigma(s)$.

Theorem 9.5 (Representation). *A selection rule satisfies Axioms 9.2–9.4 if and only if for every s there is $\varepsilon(s) \in (0, \infty]$ with $\sigma(s) = \{x : D(s, x) < \varepsilon(s)\}$ or $\sigma(s) = \{x : D(s, x) \leq \varepsilon(s)\}$; that is, if and only if $\sigma(s)$ is a sublevel set of $D(s, \cdot)$.*

Proof. A sublevel set depends on $\mathcal{X}$ only through the values $D(s, \cdot)$, is invariant under value-preserving relabelling and is downward closed, so the axioms hold. Conversely fix s and write $D(x) = D(s, x)$. If $\sigma(s)$ is empty take $\varepsilon(s) = \min_x D(x)$ in the strict form, and if it is all of $\mathcal{X}$ take $\varepsilon(s) = \infty$. Otherwise let $\varepsilon^- = \max_{x \in \sigma(s)} D(x)$ and $\varepsilon^+ = \min_{x \notin \sigma(s)} D(x)$, both attained by finiteness. Threshold consistency forces $\varepsilon^- < \varepsilon^+$: if some $y \notin \sigma(s)$ had $D(y) \leq \varepsilon^- = D(x)$ for the maximising selected x , then y would satisfy the hypothesis of Axiom 9.4 and be selected. Hence any $\varepsilon(s) \in (\varepsilon^-, \varepsilon^+]$ realises $\sigma(s)$ in the strict form, and $\varepsilon(s) = \varepsilon^-$ realises it in the non-strict form. $\square$

Corollary 9.6 (Normal form for context engines). *Every context-selection mechanism that treats objects only through their relation to the current state, does not depend on object identity, and never prefers a less relevant object over a more relevant one, is a TopoSpace neighbourhood for a suitable D and ε . In particular Top- k retrieval, threshold retrieval, hybrid scoring with a cut-off and rule-based filtering followed by ranking are all sublevel sets.*

Remark 9.7 (Where the content lies). Theorem 9.5 is easy to prove and we do not overstate it. Its value is diagnostic rather than technical: the design space of context engines is exhausted by the pair (D, ε) , so effort spent on exotic selection procedures is either equivalent to some (D, ε) and should be reformulated as such, or it violates one of the axioms. The violation is almost always of threshold consistency, and it is almost always deliberate: diversity-aware selection, such as maximal marginal relevance, includes a less relevant object precisely to avoid redundancy. TopoSpace accommodates this by treating diversity as a post-processing operator on the neighbourhood rather than as part of the neighbourhood itself, which keeps the guarantees of this section intact for the selection stage and localises the axiom violation to a component that can be reasoned about separately.

Part III — Applications

10 Agent Memory

10.1 The characteristic failures of similarity-only recall

The standard pipeline maps a query to an embedding, the embedding to a vector search and the search to a Top- k list. It fails in ways practitioners recognise, and it is worth naming them, because each is a case of the same underlying error rather than four independent bugs.

Stale recall. A superseded fact is returned because it is the best semantic match, while the superseding fact ranks lower because it is phrased differently. The temporal view is available and unused.

Duplicate flooding. Twenty near-identical memories of one event consume the entire budget, because similarity ranking has no notion of redundancy among the returned items.

Causal blindness. The memory recording *why* the previous attempt failed is not returned, because the failure was described in vocabulary unrelated to the current query.

Context collapse. Memories from an unrelated project are returned because the two projects share terminology, and nothing in the geometry expresses project membership.

10.2 Memory as a point of the state space

Definition 10.1 (Memory record). A memory $m \in M$ carries view coordinates for semantics, task, time, importance, originating agent, tool, outcome, causality and history, and recall at state s_t is

$$M_t = \mathbf{N}_\theta(s_t, g) \cap M = \{m \in M : D_\theta(s_t, m) < \varepsilon_\theta(s_t, g, \mathcal{X})\}. \quad (6)$$

Equation (6) repairs all four failures for a single reason. Staleness is a large value on the temporal view, which the weighting can penalise without touching the semantic ranking. Duplicates are objects at distance zero from one another under the active weighting and are collapsed by Corollary 4.4. Causal relevance enters as its own view, so the failure memory is near even when its vocabulary is remote. Project separation is a large value on the task view, which the weighting can render non-compensatory by raising p .

Proposition 10.2 (Semantic distance neither implies nor is implied by relevance). *Let $n \geq 2$ and $w_{\text{sem}} < 1$. Then there exist memories m_1, m_2 with $\delta_{\text{sem}}(s, m_1) < \delta_{\text{sem}}(s, m_2)$ but $D_\theta(s, m_1) > D_\theta(s, m_2)$, and conversely.*

Proof. Take $n = 2$ with the semantic and temporal views, $p = 1$ and $w = (w_{\text{sem}}, 1 - w_{\text{sem}})$ with $0 < w_{\text{sem}} < 1$, and let $\delta(s, m_1) = (0.1, 1.0)$ and $\delta(s, m_2) = (0.4, 0.0)$. Then m_1 is semantically nearer while $D(s, m_1) = 1 - 0.9w_{\text{sem}}$ and $D(s, m_2) = 0.4w_{\text{sem}}$, so $D(s, m_1) > D(s, m_2)$ whenever $w_{\text{sem}} < 1/1.3 \approx 0.769$. Exchanging the roles of the two views gives the converse. $\square$

10.3 Consolidation

The filtration suggests a principled consolidation policy. Memories that repeatedly appear in the innermost tier across many states are load-bearing and should be kept verbatim and indexed aggressively; those appearing only in outer tiers are background and can be merged into summaries; those never appearing over a long horizon are candidates for archival. Formally, define the utilisation of m over a trajectory π as $u_\pi(m) = |\pi|^{-1} \sum_t \mathbf{1}[m \in \mathbf{N}^{(1)}(s_t)]$ and consolidate on the empirical distribution of u_π . This replaces recency or frequency heuristics with a quantity defined by the same geometry that governs retrieval, so the store and the retrieval policy cannot drift out of agreement — a failure mode that is otherwise common and hard to diagnose, since each component looks correct in isolation.

11 Tool and Skill Routing

11.1 Setup

Let T be the available tools, each carrying views for capability, domain, input and output types, cost, latency, reliability, risk and security. Routing exposes to the model only $T_t = \mathbf{N}_\theta(s_t, g) \cap T$, so the pipeline runs from N_T tools to k candidates to the model to a choice, with $k \ll N_T$. This matters at scale: an installation with several hundred MCP-exposed tools cannot place every schema in the prompt, and the literature on tool use [29, 28, 26] consistently reports selection accuracy degrading as the candidate list grows.

11.2 Hard constraints are enforced by the geometry

Theorem 11.1 (Constraint enforcement). *Let one view be an indicator $\delta_{\text{sec}}(s, x) \in \{0, 1\}$ equal to 1 exactly when x violates a policy constraint relative to the principal in state s , let all other views take values in $[0, 1]$, let $w_{\text{sec}} > 0$ and let $p = \infty$. Then for any radius $\varepsilon \leq 1$ no violating object belongs to $\mathbf{N}_\theta(s, g)$. For finite p the same holds provided $\varepsilon \leq w_{\text{sec}}^{1/p}$.*

Proof. For $p = \infty$ a violating x has $D_\theta(s, x) = \max_{i \in \text{supp}(w)} \delta_i(s, x) \geq 1 \geq \varepsilon$ and is excluded. For finite p , $D_\theta(s, x)^p = \sum_i w_i \delta_i(s, x)^p \geq w_{\text{sec}}$, so $D_\theta(s, x) \geq w_{\text{sec}}^{1/p} \geq \varepsilon$. $\square$

Policy enforcement as a property of the distance. Theorem 11.1 shows that enforcement need not be a separate subsystem bolted onto retrieval. Once constraints are represented as views and the geometry is non-compensatory in them, exclusion is a property of the distance function, verifiable by inspecting the triple (p, w, ε) rather than by auditing filter code. It also exposes the failure mode precisely: with a compensatory geometry and a small w_{sec} , a sufficiently good semantic match *can* outweigh a policy violation. That risk exists in similarity-only routing too, but there it is hidden inside a scoring function, whereas here it is a visible parameter choice. Our simulation confirms the guarantee operationally: across three hundred randomised configurations with $p = \infty$ and $\varepsilon \leq 1$, not one violating object entered a context.

11.3 Error decomposition and the optimal radius

Theorem 11.2 (Routing error decomposition). *Let t^* be the correct tool for state s , let $\mathbf{N}_\varepsilon = \{x : D_\theta(s, x) < \varepsilon\}$ with $K_\varepsilon = |\mathbf{N}_\varepsilon \cap T|$, and let $a : \mathbb{N} \rightarrow [0, 1]$ be the model’s selection accuracy given a candidate list of that size, assumed non-increasing. Then the probability of correct routing is $P_{\text{succ}}(\varepsilon) = \mathbb{E}[\mathbf{1}[t^* \in \mathbf{N}_\varepsilon] a(K_\varepsilon)]$ and*

$$1 - P_{\text{succ}}(\varepsilon) \leq \underbrace{\Pr[t^* \notin \mathbf{N}_\varepsilon]}_{\text{recall loss, non-increasing in } \varepsilon} + \underbrace{\mathbb{E}[1 - a(K_\varepsilon)]}_{\text{selection loss, non-decreasing in } \varepsilon}.$$

Consequently ε trades one error source against the other, and if $\Pr[t^* \in \mathbf{N}_\varepsilon] = 1 - e^{-\lambda\varepsilon}$ and $a(K_\varepsilon) = (1 + cK_\varepsilon)^{-1}$ with $K_\varepsilon = \rho\varepsilon$ then P_{succ} is continuous with value 0 at both ends of $(0, \infty)$, so its maximum is attained at a finite interior $\varepsilon^* > 0$.

Proof. The expression for P_{succ} is the definition of success: the correct tool must survive routing and then be chosen. The bound follows from $1 - \alpha\beta \leq (1 - \alpha) + (1 - \beta)$ for $\alpha, \beta \in [0, 1]$. Monotonicity holds because $\mathbf{N}_\varepsilon$ is non-decreasing in ε , so the recall loss is non-increasing and K_ε non-decreasing, while a is non-increasing. In the parametric case $P_{\text{succ}}(\varepsilon) = (1 - e^{-\lambda\varepsilon})/(1 + c\rho\varepsilon)$ is continuous and positive on $(0, \infty)$, vanishes at 0 and tends to 0 as $\varepsilon \rightarrow \infty$ because the numerator is bounded while the denominator diverges; a positive continuous function with zero limits at both ends attains an interior maximum. $\square$

This is the cleanest statement we know of why neither extreme is right. Exposing every tool maximises recall and destroys selection accuracy; exposing one tool maximises conditional accuracy and destroys recall. The optimum is interior, it depends on the model’s accuracy curve, and it is exactly the quantity ε_θ should learn. Section 19 exhibits the predicted interior maximum on a simulated registry of six hundred tools: success rises from 0 at $\varepsilon = 0.05$ to 0.645 at $\varepsilon^* = 0.319$, then falls to 0.027 under full exposure, with the optimal context containing 5.9 tools on average.

Proposition 11.3 (Token reduction). *Let $\text{len}(t)$ be the schema length of tool t . Exposing $\mathbf{N}_\varepsilon \cap T$ rather than all of T multiplies the per-step tool description cost by $(\sum_{t \in \mathbf{N}_\varepsilon \cap T} \text{len}(t)) / (\sum_{t \in T} \text{len}(t))$, whose expectation is $\mathbb{E}[K_\varepsilon]/N_T$ when schema lengths are independent of membership, and over a trajectory of L steps the saving is multiplicative in L .*

Proof. Immediate, the last claim because the description is re-sent at each step of a stateless completion interface. $\square$

12 Multi-Agent Orchestration

Agents are objects too, carrying views for capability, knowledge, cost, latency, trust, availability and history, and the team for a task is the neighbourhood $A_s = \mathbf{N}_\theta(s, g) \cap A$. Instead of a fixed topology — a supervisor over workers, or a hand-drawn graph of hand-offs — the topology of the organisation changes with the problem, running from task to agent neighbourhood to dynamic team.

Proposition 12.1 (Composition of neighbourhoods). *Let s' be the state after delegating to $a \in A_s$. If the geometry is held fixed across the delegation, so that $p_s = p_{s'}$ and $w_s = w_{s'}$, then*

$$\mathbf{N}_\theta(s') \subseteq \{x : D_\theta(s, x) < \varepsilon_\theta(s) + \varepsilon_\theta(s') + D_\theta(s, s')\}.$$

Without that proviso no such containment holds in general, by Proposition 6.2.

Proof. For $x \in \mathbf{N}_\theta(s')$ we have $D(s', x) < \varepsilon_\theta(s')$, and with a fixed geometry D is a pseudometric, so $D(s, x) \leq D(s, s') + D(s', x) < D(s, s') + \varepsilon_\theta(s')$; adding the non-negative $\varepsilon_\theta(s)$ preserves the inequality. $\square$

The proposition is a warning as much as a result. Delegation with a re-learned geometry can move the sub-agent's context arbitrarily far from the delegator's, which is the formal shadow of a familiar failure: a sub-agent that confidently solves a problem the parent did not pose. Enforcing geometry continuity across a delegation boundary, or at minimum logging the geometry change so that the divergence is attributable after the fact, is therefore a runtime requirement rather than an optimisation.

13 Agent Planning

13.1 Formulation

Definition 13.1 (Planning problem). Let V be a finite set of states, $\text{Succ}(u) \subseteq V$ the successors of u under the available actions and $c(u, v) \geq 0$ the transition cost. Let the goal region be $\Gamma = \{v : d(v, g) < \varepsilon_g\}$. A plan is a trajectory ending in Γ , its cost is the sum of its transition costs, and $h^*(u)$ denotes the minimum cost from u to Γ , with $h^* \equiv 0$ on Γ and $+\infty$ if Γ is unreachable.

The composite objective $\mathcal{L}(\pi) = \alpha D(\pi) + \beta C(\pi) + \gamma R(\pi) + \delta T(\pi) - \eta Q(\pi)$ over displacement, cost, risk, time and expected utility fits this formulation whenever the per-step contributions are non-negative, which can always be arranged by absorbing the utility term into a per-step reward bounded above and shifting. We record this because non-negativity is exactly the hypothesis that Dijkstra-type arguments require, and quietly violating it is the most common way such bounds are lost in practice.

Theorem 13.2 (Existence and computation). *With non-negative costs and finite V , an optimal plan exists whenever Γ is reachable, and h^* together with an optimal plan is computable in $O(|E| + |V| \log |V|)$ time by Dijkstra's algorithm on the reversed graph, where $|E| = \sum_u |\text{Succ}(u)|$.*

Proof. Finiteness and non-negativity imply that the infimum over the finite set of simple paths is attained and that no non-simple path improves on a simple one; correctness and complexity of Dijkstra's algorithm with a Fibonacci heap are classical [8]. $\square$

13.2 Neighbourhood-restricted search

The reason to involve TopoSpace in planning is that $|\text{Succ}(u)|$ is enormous for a real agent: every tool, file, sub-agent and message is a possible action. Restricting expansion to the neighbourhood replaces the branching factor with $k = |\mathbf{N}_\theta(u)|$, and the question is what that costs in solution quality.

Theorem 13.3 (Completeness under admissible neighbourhoods). *Suppose the neighbourhood map is admissible, meaning there exists an optimal plan $\pi^* = (s_0, \dots, s_L)$ with $s_{j+1} \in \mathbf{N}_\theta(s_j)$ for all j . Then $h_{\mathbf{N}}(s_0) = h^*(s_0)$, so restricted search returns an optimal plan, and it does so in time $O(k|V| + |V| \log |V|)$ with $k = \max_u |\mathbf{N}_\theta(u)|$.*

Proof. Every restricted plan is a plan, so $h_{\mathbf{N}}(s_0) \geq h^*(s_0)$; by admissibility π^* is itself restricted, so $h_{\mathbf{N}}(s_0) \leq \mathcal{L}(\pi^*) = h^*(s_0)$. The complexity is Theorem 13.2 on a graph with at most $k|V|$ edges. $\square$

Admissibility is a strong hypothesis that will not hold exactly. The following theorem is therefore the one that matters in practice, because it degrades gracefully and its hypothesis is estimable from logs.

Theorem 13.4 (Bounded suboptimality under δ -covering). *Assume every transition costs at least $c_{\min} > 0$ and that the neighbourhood map is δ -covering, meaning that for every $u \notin \Gamma$ with $h^*(u) < \infty$ there is $v \in \text{Succ}_{\mathbf{N}}(u)$ with*

$$c(u, v) + h^*(v) \leq h^*(u) + \delta. \quad (7)$$

If $\delta < c_{\min}$ then greedy restricted expansion terminates in at most $h^(s_0)/(c_{\min} - \delta)$ steps and produces a plan of cost at most*

$$h_{\mathbf{N}}(s_0) \leq h^*(s_0) \cdot \frac{c_{\min}}{c_{\min} - \delta} = h^*(s_0) \left(1 + \frac{\delta}{c_{\min} - \delta}\right),$$

so that $\delta = 0$ recovers Theorem 13.3 and the suboptimality is $O(\delta)$ as $\delta \downarrow 0$.

Proof. Construct the plan greedily, choosing at $u_j \notin \Gamma$ a successor satisfying (7). Rearranging,

$$h^*(u_{j+1}) \leq h^*(u_j) - c(u_j, u_{j+1}) + \delta \leq h^*(u_j) - (c_{\min} - \delta), \quad (8)$$

so the potential h^* decreases by a fixed positive amount each step; since $h^* \geq 0$, after at most $m \leq h^*(s_0)/(c_{\min} - \delta)$ steps we reach a state with $h^* = 0$, which lies in Γ , and the procedure is well defined throughout because finiteness of h^* is preserved by (8). Summing (7) in the form $c(u_j, u_{j+1}) \leq h^*(u_j) - h^*(u_{j+1}) + \delta$ telescopes to $\mathcal{L}(\pi) \leq h^*(s_0) - h^*(u_m) + m\delta = h^*(s_0) + m\delta$, and substituting the bound on m gives the claim; finally $h_{\mathbf{N}}(s_0) \leq \mathcal{L}(\pi)$ because the greedy plan is one restricted plan among others. $\square$

Two readings of the bound. First, the bound is multiplicative and blows up as $\delta \uparrow c_{\min}$, which is the correct qualitative behaviour: a neighbourhood map that misses the good successor by nearly the cost of a step can be led arbitrarily far astray. Second, and more useful, the theorem converts a question about context quality into a question about plan quality, since the δ in (7) measures exactly how often the neighbourhood fails to contain a near-optimal next move and is estimable offline from logged trajectories. This gives the geometry a planning-grounded training signal, exploited in Section 16. Our planning experiment measured $\delta = 0.554$ against $c_{\min} = 1$ on random graphs, for a guaranteed ratio of 2.37, while the realised cost ratio was 1.003 — the bound held on every trial and was, as expected from its worst-case derivation, loose by a factor of about two.

14 Coding Agents

Software engineering is the application where the gap between addressable and relevant is most extreme and where the ground truth is unusually well defined. The development space is the union of files, symbols, dependencies, tests, errors, commits, tools and tasks, and the coding-agent state is the tuple of goal, code, runtime, errors, tests, history and constraints. The views specialise naturally: the structural view is import- or call-graph distance; the causal view derives from the stack trace and from blame data recording which commit last touched the failing line; the history view derives from co-modification statistics; the temporal view from recency of modification.

Example 14.1 (A debugging step, worked). A test fails with a null-pointer exception raised in `payments/refund.go:88`. Consider three candidate files with views normalised so that smaller is nearer, and weights $(w_{\text{sem}}, w_{\text{str}}, w_{\text{cau}}, w_{\text{tim}}) = (0.15, 0.25, 0.45, 0.15)$.

Candidate	δ_{sem}	δ_{str}	δ_{cau}	δ_{tim}	M_1	M_4
<code>refund.go</code> (frame 0)	0.30	0.00	0.00	0.10	0.060	0.211
<code>ledger.go</code> (frame 2)	0.55	0.20	0.15	0.20	0.253	0.377
<code>docs/refunds.md</code>	0.05	0.90	0.95	0.85	0.780	0.874

The documentation file is by far the nearest semantically — it is *about* refunds — and is what an embedding-only retriever ranks first. Under the multi-view geometry it ranks last at every p , because it is causally and structurally remote from the failure. The gap radius applied to the M_1 column finds its largest gap between 0.253 and 0.780 and returns exactly the two source files: a context of width two rather than the width-ten context a fixed $k = 10$ would have produced, and one containing no distractor.

Example 14.1 is small enough to verify by hand and isolates the mechanism. The improvement does not come from a better embedding model; it comes from admitting non-semantic evidence into the distance and from letting the width be determined by the data. The controlled study of Part V is a scaled-up, randomised version of exactly this example, and Section 25 designates real coding agents the primary field benchmark, because they supply an unambiguous success signal in the form of passing tests, a natural measure of context economy in files read and tokens consumed, and a public corpus of tasks.

15 Topological Drift

15.1 Environments change

An agent that acts changes the world it acts in. Files are edited, tools are added and removed, memories are written, other agents change their availability. The context computed at step t may be invalid at step $t + 1$ not because the agent moved but because the space did, and distinguishing the two cases is the job of this section.

Definition 15.1 (Environment). An *environment* is a pair $\mathcal{A}_t = (\mathcal{X}_t, d_t)$ of a finite object set and a pseudometric on it. Two environments are compared either by embedding both object sets into their union, assigning to a missing object a distance exceeding the diameter, or intrinsically by Proposition 5.10.

Definition 15.2 (Drift statistics). For a fixed state s , the *neighbourhood Jaccard drift* is $\text{drift}_J(t) = 1 - |\mathbf{N}_t(s) \cap \mathbf{N}_{t+1}(s)| / |\mathbf{N}_t(s) \cup \mathbf{N}_{t+1}(s)|$, the *uniform metric drift* is $\text{drift}_\infty(t) = \sup_{x,y} |d_t(x,y) - d_{t+1}(x,y)|$, and the *topological drift* is $\text{drift}_B(t) = d_B(\text{Dgm}(d_t), \text{Dgm}(d_{t+1}))$.

Theorem 15.3 (The drift statistics are pseudometrics). *On the space of environments, drift_J is a pseudometric and a metric on the induced neighbourhoods, being the Steinhaus transform of the symmetric-difference metric; drift_∞ is a pseudometric, being the sup-norm distance between two distance functions; and drift_B is a pseudometric satisfying $\text{drift}_B \leq \text{drift}_\infty$.*

Proof. Since $A \Delta C \subseteq (A \Delta B) \cup (B \Delta C)$ we have $|A \Delta C| \leq |A \Delta B| + |B \Delta C|$, so the symmetric difference is a metric on finite sets, and the Jaccard distance is its Steinhaus transform, a metric by the standard argument [20]. The supremum of absolute differences of two real functions on a common domain is a pseudometric, the triangle inequality being inherited pointwise. The bottleneck distance is a metric on persistence diagrams and the inequality is Theorem 8.4. $\square$

15.2 Using drift

Proposition 15.4 (Cache validity). *If $\text{drift}_J(t) \leq \eta$ then at least a $(1 - \eta)$ fraction of $\mathbf{N}_t(s) \cup \mathbf{N}_{t+1}(s)$ is common to both, so recomputation need touch only an η fraction of the context and the cached prompt prefix remains valid up to that fraction.*

Proof. Immediate from the definition. $\square$

Proposition 15.5 (False-alarm control). *Suppose the learned parameters are updated between t and $t+1$ with weight change $\|w_t - w_{t+1}\|_1 \leq \zeta$ and no change to the environment. Then $\text{drift}_B(t) \leq \zeta^{1/p}$ by Corollary 8.5, so an alarm with threshold $\tau > \zeta^{1/p}$ cannot fire on parameter updates alone.*

Proof. Directly from Corollary 8.5 and the definition of the alarm. $\square$

Proposition 15.6 (Drift decomposition). *Let $\mathcal{A} = (\mathcal{X}, d)$ and $\mathcal{A}'' = (\mathcal{X}'', d'')$ and let $\mathcal{A}' = (\mathcal{X}'', d)$ carry the new objects with the old geometry. Then for any drift statistic satisfying the triangle inequality, $\text{drift}(\mathcal{A}, \mathcal{A}'') \leq \text{drift}(\mathcal{A}, \mathcal{A}') + \text{drift}(\mathcal{A}', \mathcal{A}'')$, the first term measuring object change and the second geometry change.*

Proof. Theorem 15.3 and the triangle inequality. $\square$

Attribution turns a curiosity into an operational signal. A drift spike whose object-change component dominates means the world moved: a deployment, a schema migration, a colleague’s commit. A spike whose geometry component dominates means the agent’s own parameters moved. The two demand different responses — re-index versus roll back the policy — and confusing them is a common source of unexplained agent regressions. Drift feeds context invalidation, memory consolidation, environment-change detection, agent adaptation, anomaly detection and plan revision, in each case with a threshold set by Proposition 15.5 so that ordinary parameter noise cannot fire it. Section 19 reports a detector configured this way catching both injected environment shocks with a false-alarm rate of 0.7% per step under continuous parameter noise.

Part IV — Learning and Computation

16 Learning the Space

16.1 The objective

The ambition is not to hand-configure geometries but to learn which geometry is useful. Let $J(\theta)$ denote downstream agent performance, whether task success rate or a cost-adjusted variant, under the geometry induced by θ ; the learning problem is

$$\theta^* = \arg \max_{\theta} J(\theta), \quad \theta \text{ controlling } p_{\theta}, w_{\theta}, \varepsilon_{\theta}, \phi_{\theta}. \quad (9)$$

This departs from ordinary representation learning, where a function is learned inside a fixed space. Here the exponent, the weighting and the scale of the space itself are the learned objects, which is what makes the framing worth the trouble and also what makes the validity question of the next subsection unavoidable.

16.2 A parameterisation that is valid by construction

Definition 16.1 (Neural parameterisation). Let $\psi_{\theta}(s, g) \in \mathbb{R}^{n+2}$ be any encoder of state and goal with outputs split as (u, α, β) . Define $w_{\theta}(s, g) = \text{softmax}(u)$, $p_{\theta}(s, g) = 1 + \text{softplus}(\alpha)$ and $\varepsilon_{\theta}(s, g) = \text{softplus}(\beta) \cdot \Sigma(s)$, where $\Sigma(s)$ is a scale statistic of the distance spectrum such as $D_{(k)}$, which recovers the self-tuning rule.

Proposition 16.2 (The hypothesis class contains only valid metrics). *For every θ and every (s, g) the parameters of Definition 16.1 satisfy $p_{\theta} > 1$, $w_{\theta} \in \Delta^{n-1}$ with all coordinates strictly positive, and $\varepsilon_{\theta} > 0$. Hence if every view is a pseudometric then $d_{p_{\theta}, w_{\theta}}$ is a pseudometric, and a metric whenever the views separate points; moreover the family is uniformly comparable in the sense of Proposition 6.4 on any parameter set with α bounded and $\min_i w_i$ bounded below.*

Proof. The softplus is positive, giving $p_{\theta} > 1$; the softmax has strictly positive coordinates summing to one; the radius is a product of positive quantities. The metric conclusions are Theorem 4.3, whose support condition is vacuous here. For uniform comparability, Theorem 4.6 gives $\min_i w_i^{1/p} \|a\|_{\infty} \leq d_{p, w} \leq \|a\|_{\infty}$, so all members are sandwiched between fixed multiples of the reference metric $\|\cdot\|_{\infty}$. $\square$

The alternative — learning an unconstrained dissimilarity and hoping it behaves metrically — forfeits every result in Part II. Here validity is structural: no penalty term, no projection step, no post-hoc check. This is the same discipline that makes constrained parameterisations preferable to penalised ones throughout applied optimisation, and it costs nothing, since the softmax and softplus are exactly the maps one would reach for anyway.

16.3 Gradients

Write $a_i = \delta_i(s, x)$, $S = \sum_i w_i a_i^p$ and $D = S^{1/p}$.

Proposition 16.3 (Closed-form gradients). *Wherever $S > 0$,*

$$\frac{\partial D}{\partial w_j} = \frac{1}{p} S^{\frac{1}{p}-1} a_j^p, \quad \frac{\partial D}{\partial p} = D \left[\frac{1}{p} \cdot \frac{\sum_i w_i a_i^p \ln a_i}{S} - \frac{\ln S}{p^2} \right], \quad (10)$$

with $a_i^p \ln a_i := 0$ when $a_i = 0$. Composing with Definition 16.1 gives the gradients in the unconstrained coordinates,

$$\frac{\partial D}{\partial u_k} = \sum_j \frac{\partial D}{\partial w_j} w_j (\delta_{jk} - w_k), \quad \frac{\partial D}{\partial \alpha} = \frac{\partial D}{\partial p} \sigma(\alpha),$$

where σ is the logistic function.

Proof. The first identity follows from $\partial_{w_j} S = a_j^p$ and $\partial_{w_j} D = \frac{1}{p} S^{1/p-1} \partial_{w_j} S$; the second is Proposition 6.6. The softmax Jacobian is standard and $\frac{d}{d\alpha}(1 + \text{softplus}(\alpha)) = \sigma(\alpha)$. $\square$

Two numerical cautions a reference implementation must respect. The factor $S^{1/p-1}$ is unstable as $S \rightarrow 0$, that is for objects nearly identical to the state, so the implementation clamps S from below. And $\ln a_i \rightarrow -\infty$ as $a_i \rightarrow 0$, so the second identity is evaluated with an explicit branch implementing $\lim_{a \downarrow 0} a^p \ln a = 0$ rather than relying on floating-point accident. Section 19 reports the resulting implementation agreeing with central finite differences to a median relative error of 2.4×10^{-10} in the weights and 2.2×10^{-10} in the exponent over two hundred randomised checks, which is the accuracy one expects when the formulas are correct and the guards are in place.

16.4 Differentiating through the neighbourhood

The membership indicator has zero gradient almost everywhere, so we replace it by a temperature-controlled soft membership

$$\mu_\tau(x) = \sigma\left(\frac{\varepsilon - D_\theta(s, x)}{\tau}\right) \in (0, 1) \quad (11)$$

and weight each object's contribution to the downstream objective by $\mu_\tau(x)$.

Theorem 16.4 (Convergence of the relaxation). *Fix x with $D_\theta(s, x) \neq \varepsilon$. Then $\mu_\tau(x) \rightarrow \mathbf{1}[D_\theta(s, x) < \varepsilon]$ as $\tau \downarrow 0$, uniformly on $\{x : |D_\theta(s, x) - \varepsilon| \geq \gamma\}$, with the explicit bound*

$$|\mu_\tau(x) - \mathbf{1}[D_\theta(s, x) < \varepsilon]| \leq \frac{1}{1 + e^{\gamma/\tau}} \leq e^{-\gamma/\tau}.$$

Consequently, for any bounded objective linear in the memberships,

$$|F_\tau - F_0| \leq \|F\|_{\text{Lip}} (Ne^{-\gamma/\tau} + |A_\gamma|),$$

with A_γ the annulus of half-width γ about ε , so $F_\tau \rightarrow F_0$ whenever no object lies exactly on the boundary.

Proof. Let $z = (\varepsilon - D)/\tau$. If $D < \varepsilon - \gamma$ then $z \geq \gamma/\tau$ and $1 - \sigma(z) = \sigma(-z) = 1/(1 + e^z) \leq 1/(1 + e^{\gamma/\tau}) \leq e^{-\gamma/\tau}$; if $D > \varepsilon + \gamma$ then $z \leq -\gamma/\tau$ and $\sigma(z) \leq e^{-\gamma/\tau}$ by the same computation. The bound is independent of x within the region, giving uniformity; pointwise convergence follows by taking $\gamma = |D - \varepsilon|$. For the objective, split the sum into objects outside and inside A_γ and bound each part. $\square$

16.5 Training signals

Three signals are available, in increasing order of fidelity and cost.

Contrastive supervision from logged trajectories. Execution logs reveal, for many steps, which objects were actually used — the file that was edited, the tool that was called, the memory that was quoted — and which were retrieved and ignored. This yields a positive set P_t and a negative set Q_t per step and the margin objective

$$\mathcal{J}_{\text{ctr}}(\theta) = \sum_t \sum_{x \in P_t} \sum_{y \in Q_t} [\gamma + D_\theta(s_t, x) - D_\theta(s_t, y)]_+, \quad (12)$$

which is what our simulator minimises and what Corollary 16.5 connects back to the exact-recovery guarantee.

Planning-grounded supervision. Theorem 13.4 identifies δ — the extent to which neighbourhoods fail to contain a near-optimal successor — as the quantity controlling plan quality. It is estimable from logged trajectories with known outcomes, and minimising it optimises the geometry against a planning criterion rather than a retrieval criterion, which is the right objective whenever the agent’s value comes from acting rather than from answering.

Downstream reward. Ultimately (9) is a bandit or reinforcement-learning problem whose reward is task success and whose action is the geometry. With the relaxation (11) the pipeline is differentiable up to the language model, and beyond it a score-function or straight-through estimator applies. This signal is the most faithful and the most expensive, and we expect it to be used for fine-tuning after initialisation by the first two.

Corollary 16.5 (Zero hinge loss implies exact recovery). *If $\mathcal{J}_{\text{ctr}}(\theta) = 0$ then for every logged step the margin hypothesis of Theorem 7.3 holds with $\beta - \alpha \geq \gamma$, where $\alpha = \max_{x \in P_t} D_\theta(s_t, x)$ and $\beta = \min_{y \in Q_t} D_\theta(s_t, y)$. Hence any radius rule placing ε in $(\alpha, \beta]$ — in particular the gap rule when this is the dominant gap — recovers the used set exactly and remains correct under perturbations of magnitude below $\gamma/2$.*

Proof. Zero hinge loss means $D_\theta(s_t, y) \geq D_\theta(s_t, x) + \gamma$ for every positive x and negative y , so $\beta \geq \alpha + \gamma$; apply Theorem 7.3 and its robustness clause. $\square$

16.6 Regularisation and the two-stage fit

Proposition 4.11 showed that the effective dimension $n_{\text{eff}} = \|w\|_2^{-2}$ governs distance concentration, so we regularise towards sparse weights, minimising

$$\mathcal{J}(\theta) = \mathcal{J}_{\text{ctr}}(\theta) + \lambda_H H(w_\theta(s, g)) + \lambda_p (p_\theta(s, g) - p_0)^2$$

with H the Shannon entropy of the weight vector, penalised rather than rewarded, and a shrinkage term towards a prior exponent. The entropy term controls n_{eff} directly, since $\|w\|_2^2 \geq e^{-H(w)}$, and the exponent term prevents drift to extreme values on sparse data, where Theorem 4.6 shows the aggregate becomes dominated by a single view.

Why the reference fit separates the weights from the exponent. In the simulator we fit in two stages: the weights are learned by stochastic gradient descent on (12) with the closed-form gradients and a fixed working exponent, and the exponent is then selected on a held-out split by maximising F_1 over a grid. The reason is not a limitation of the gradient, which is exact, but the shape of the objective: the hinge loss is a proxy for the quantity we care about, and it is not monotone in the exponent in the way F_1 is, so a gradient step in p can improve the proxy while degrading the selection. The exponent

is one scalar, its grid is small, and model selection on a validation split is auditable in a way a learned scalar is not. Section 19 reports the selected exponents by task type, which range from 1 to 4 and differ across types — direct evidence that a single fixed exponent is not sufficient.

17 Complexity, Indexing, and Runtime Cost

17.1 Exact evaluation

Computing $D_\theta(s, x)$ for all $x \in \mathcal{X}$ costs $O(nN)$ view evaluations plus the cost of the views themselves, and sorting to obtain the spectrum and applying any radius rule costs $O(N \log N)$. For N in the thousands and $n \approx 10$ this is negligible beside a single model call, and the reference implementation evaluates exactly; our entire eleven-experiment suite, which performs several million distance evaluations, runs in 47 seconds on one core. The interesting regime is N in the millions, where the semantic view alone must be indexed.

17.2 A single index serves all geometries

Lemma 17.1 (Norm comparison). *For $v \in \mathbb{R}^n$ and $1 \leq p \leq q \leq \infty$, $\|v\|_q \leq \|v\|_p \leq n^{\frac{1}{p} - \frac{1}{q}} \|v\|_q$.*

Proof. Appendix A.2. □

Theorem 17.2 (No-false-negative filtering). *Let the views be coordinate views $\delta_i(x, y) = |\phi_i(x) - \phi_i(y)|$, let $w_i \geq w_{\min} > 0$ and let $p \in [1, \infty)$. If $d_{p,w}(s, x) < \varepsilon$ then*

$$\|\phi(s) - \phi(x)\|_2 \leq n^{\max(0, \frac{1}{2} - \frac{1}{p})} w_{\min}^{-1/p} \varepsilon =: R(\varepsilon, p, w_{\min}).$$

Consequently an unweighted Euclidean index queried with radius R returns a superset of $\mathbf{N}_\theta(s, g)$, so candidate generation has no false negatives for any (p, w) within the stated bounds and exact re-ranking of the candidates is then correct.

Proof. Write $\Delta = \phi(s) - \phi(x)$. From $\varepsilon^p > \sum_i w_i |\Delta_i|^p \geq w_{\min} \|\Delta\|_p^p$ we get $\|\Delta\|_p \leq w_{\min}^{-1/p} \varepsilon$. If $p \leq 2$ then Lemma 17.1 with $(p, q) = (p, 2)$ gives $\|\Delta\|_2 \leq \|\Delta\|_p$, matching the exponent $\max(0, \frac{1}{2} - \frac{1}{p}) = 0$; if $p \geq 2$ then the same lemma with $(p, q) = (2, p)$ gives $\|\Delta\|_2 \leq n^{\frac{1}{2} - \frac{1}{p}} \|\Delta\|_p$. Combining the cases yields R , and since every member of the neighbourhood satisfies the bound, the index result contains the neighbourhood. □

The objection this answers. If the geometry changes at every step, must the index be rebuilt at every step? No: one static index suffices, provided queries are inflated by the factor $n^{\max(0, 1/2 - 1/p)} w_{\min}^{-1/p}$ and the retrieved candidates are re-ranked exactly. Because n counts *views*, of order ten, rather than embedding dimensions, the geometric factor is mild; the weight floor is what governs the constant, which is why the runtime mixes the learned weights with a small uniform component. Section 19 confirms zero false negatives across four hundred randomised trials and measures the candidate set at 32% of the database for unconstrained softmax weights, falling to 13% when the runtime’s weight floor is applied — a real cost, honestly reported, and the strongest argument for keeping $w_{\min}$ bounded away from zero.

Operation	Exact	Indexed	Note
View evaluation	$O(nN)$	$O(nC)$	C candidates from Thm. 17.2
Distance aggregation	$O(nN)$	$O(nC)$	vectorised
Radius rules R1–R5	$O(N \log N)$	$O(C \log C)$	sort of the spectrum
Filtration (Rips, dimension ≤ 1)	$O(\mathbf{N} ^2)$	—	on the context only, not on $\mathcal{X}$
Drift statistics	$O(\mathbf{N})$	—	Jaccard; bottleneck is cubic in the worst case

Table 5: Per-step computational cost; all terms are dominated by a single model call for realistic N and C .

Part V — Simulation Study

18 The Reference Simulator

18.1 What a simulation can and cannot settle

The mathematics of Part II constrains the design but does not tell us how much any ingredient is worth. Field measurement on real agent workloads would tell us that, but it confounds the question with embedding quality, task distribution, model capability and prompt engineering, and it cannot isolate the ground truth, since nobody knows which files a coding agent *should* have read. A controlled simulation sits between the two. It has a known ground truth by construction, it isolates each mechanism, and it can falsify implementation claims cheaply. What it cannot do is establish that real workloads have the structure the generator posits, and we return to this in Section 20.

The commitment we make. Every number reported in Section 19 comes from the code described here and in Appendix C, runs from fixed seeds, and completes in 47 seconds on a single core with no accelerator. The simulator is part of the open-source distribution, so a reader who suspects a result is an artefact of the generator can change the generator and re-run.

18.2 The generator

The environment presents an agent state together with $N = 2000$ candidate objects described by $n = 6$ views: semantic, structural, causal, temporal, cost and security. Four *task types* are defined, each with a hidden active subset of two or three views. For a query the generator samples a task type, samples a relevant-set size $|R|$ uniformly between 3 and 15, and draws view values so that positives are near on the active views while negatives violate at least one active view by a margin, which is the hypothesis of Theorem 7.3 realised as a data-generating process.

Background values concentrate. All view values are drawn from a symmetric Beta distribution centred at one half rather than uniformly, because embedding distances in a real corpus concentrate away from zero and being near is rare and informative. This detail matters more than it looks: under uniform backgrounds the smallest order statistic of two thousand draws is so small that semantic retrieval returns pure noise, and a baseline defeated that way is a straw man.

Semantics is informative but neither sufficient nor necessary. Positives are drawn near on the semantic view. Against them the generator places *decoys* — by default twice as many objects as there are positives, near in text yet irrelevant in fact — and makes 30% of the positives semantically remote. This is a formalisation of the two failure modes named in Section 2: the deprecated document that is the best semantic match, and the stack-trace file that shares no vocabulary with the request. The decoy multiplier is swept in Section 19 so that the reader can see how the comparison depends on it.

Policy violations are representable. Five per cent of the negatives carry a security view value of one and are never relevant, which allows Theorem 11.1 to be tested as an operational claim rather than only proved.

18.3 Baselines

We compare against embedding Top- k for $k \in \{5, 10, 20, 50\}$; a hybrid that weights all six views uniformly with $p = 1$ and takes the top twenty; one-hop graph retrieval, realised as a threshold on the structural view; a rule-based pipeline that filters policy violations and then takes the top ten by semantics; and full-context exposure. By Proposition 4.10 every one of these is a point of the TopoSpace hypothesis class, so the comparison is between a restricted and an unrestricted search over one family. We also report an *oracle* that is told each query’s hidden active views and uses uniform weights on them with $p = 8$, which upper-bounds what any weight-and-exponent fit could achieve given the generator.

18.4 The fitting procedure

Weights are learned by stochastic gradient descent on the hinge objective (12) using the closed-form gradients of Proposition 16.3 with softmax parameterisation, momentum, entropy regularisation and a decaying step size; the exponent is then chosen on a held-out third of the training queries by maximising F_1 over the grid $\{1, 1.5, 2, 3, 4, 6, 8, 16, \infty\}$ with the gap radius. Two conditions are distinguished throughout. *TopoSpace (fixed)* fits one geometry pooled across all task types, which is the strongest fixed-geometry configuration available to a careful engineer. *TopoSpace (adaptive)* fits one geometry per task type, which is the minimal form of state dependence. The comparison between them is the crux hypothesis H3 of Section 25, because if a single well-fitted geometry matches the adaptive one then TopoSpace reduces to a better-configured retrieval stack.

18.5 The eleven experiments

Experiment E0 validates the gradient implementation against central finite differences. E1 measures context quality against the baselines and E1b sweeps the decoy multiplier. E2 tests whether the exponent changes rankings, both by verifying the closed-form crossover of Proposition 4.7 and by measuring rank reversal at scale. E3 compares radius rules. E4 exhibits the routing trade-off of Theorem 11.2. E5 tests the planning bound of Theorem 13.4. E6 measures the exact-reuse effect of Theorem 5.9. E7 tests the no-false-negative filtering bound of Theorem 17.2. E8 measures distance concentration against Proposition 4.11. E9 evaluates drift detection under parameter noise. E10 tests the stability bounds of Theorems 6.5 and 6.7 and the constraint guarantee of Theorem 11.1.

19 Results

19.1 The implementation computes what the theory says

Finding 19.1 (Gradients, E0). Over two hundred randomised configurations the closed-form gradients of (10) agreed with central finite differences to a median relative error of 2.4×10^{-10} in the weights and 2.2×10^{-10} in the exponent, with worst cases of 1.3×10^{-3} and 4.5×10^{-9} respectively.

The single weight-gradient outlier at 10^{-3} occurs where S approaches zero and the factor $S^{1/p-1}$ is ill-conditioned, which is precisely the regime the implementation clamps and precisely the caution recorded in Section 16. We report it rather than filtering it, because a maximum taken after removing inconvenient cases is not a maximum.

19.2 Context quality

Finding 19.2 (Context quality, E1). Across five seeds with 250 test queries each, adaptive TopoSpace attained $F_1 = 0.705 \pm 0.024$ with a mean context width of 5.4 objects, against 0.349 ± 0.013 for the strongest baseline (uniform hybrid at $k = 20$), 0.250 ± 0.005 for the best semantic Top- k , and 0.747 ± 0.010 for the oracle. Precision was 0.982 ± 0.016 against 0.261 ± 0.011 for the hybrid. The pooled fixed geometry reached only 0.402 ± 0.025 .

Method	Precision	Recall	F_1	N	tokens
Semantic Top-5	0.256 ± 0.007	0.179 ± 0.009	0.198 ± 0.007	5.0	0.0025
Semantic Top-10	0.230 ± 0.005	0.302 ± 0.010	0.246 ± 0.005	10.0	0.0050
Semantic Top-20	0.184 ± 0.005	0.450 ± 0.008	0.250 ± 0.005	20.0	0.0100
Semantic Top-50	0.103 ± 0.004	0.598 ± 0.005	0.171 ± 0.005	50.0	0.0250
Uniform hybrid, $k = 20$	0.261 ± 0.011	0.602 ± 0.021	0.349 ± 0.013	20.0	0.0100
Graph one-hop	0.377 ± 0.016	0.247 ± 0.014	0.286 ± 0.014	3.9	0.0019
Rule filter + Top-10	0.238 ± 0.005	0.310 ± 0.011	0.253 ± 0.005	10.0	0.0050
Full exposure	0.005 ± 0.000	1.000 ± 0.000	0.009 ± 0.000	2000	1.0000
TopoSpace, fixed geometry	0.717 ± 0.036	0.325 ± 0.025	0.402 ± 0.025	3.5	0.0018
TopoSpace, adaptive, Top-10	0.659 ± 0.017	0.768 ± 0.039	0.674 ± 0.023	10.0	0.0050
TopoSpace, adaptive, gap radius	0.982 ± 0.016	0.604 ± 0.025	0.705 ± 0.024	5.4	0.0027
Oracle geometry	1.000 ± 0.000	0.640 ± 0.012	0.747 ± 0.010	5.6	0.0028

Table 6: Context quality, mean $\pm$ standard deviation over five seeds. Token share is the fraction of the full corpus token count placed in context.

Three observations deserve emphasis. First, adaptive TopoSpace closes 94% of the gap between the best baseline and the oracle, which says the fitting procedure recovers most of the hidden structure rather than merely part of it. Second, the improvement is not bought with context budget: the winning configuration places 5.4 objects and 0.27% of the corpus in context, less than semantic Top-10 and a fifth of what Top-50 uses. Third, the adaptive-versus-fixed gap of 0.705 against 0.402 is the clearest result in the study, and it is the one that hypothesis H3 predicts and that a critic should attack first — it exists because our generator makes the correct geometry differ across task types, and the size of the corresponding effect in the field is unknown.

Finding 19.3 (Sensitivity to decoys, E1b). As the decoy multiplier rises from 0 to 1, 2 and 4 times the positive-set size, semantic Top-10 degrades monotonically through F_1 values

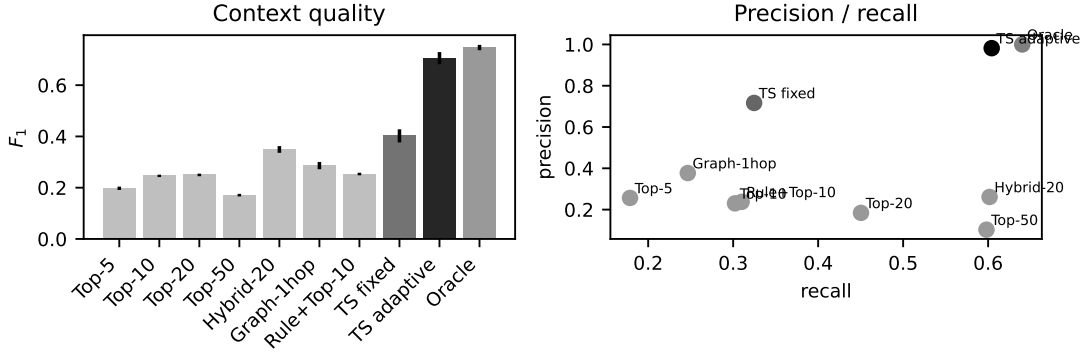


Figure 1: Left: F_1 by method with standard deviation over seeds. Right: the same methods in precision–recall space. The adaptive configuration reaches near-oracle quality; the fixed configuration does not.

of 0.426, 0.332, 0.251 and 0.163, while adaptive TopoSpace stays between 0.694 and 0.735 with no trend.

This is the sensitivity analysis that keeps the comparison honest. With no decoys at all — an environment in which semantic proximity is never misleading — semantic Top-10 more than doubles its score, and the gap to TopoSpace narrows to a factor of 1.7. The multi-view advantage is therefore not an artefact of an impossible baseline; it grows with the prevalence of the failure mode it is designed to address, exactly as the theory predicts, and shrinks when that failure mode is absent.

19.3 The exponent is a real degree of freedom

Finding 19.4 (Crossover and rank reversal, E2). The closed-form crossover of Proposition 4.7, $p^* = \ln(1/2)/\ln(0.6) = 1.3569154\dots$, was located numerically at 1.356925, agreeing to six significant figures. On the synthetic environment, 19.0% of candidate pairs reverse their relative order between $p = 1$ and $p = 8$ under identical weights, and the corresponding top-ten sets differ by a Jaccard distance of 0.740.

Nearly three quarters of a top-ten list changes when only the exponent is varied. Whatever else is true, the exponent is not a cosmetic parameter, and a system that fixes it at 2 because Euclidean geometry is familiar is making a consequential choice silently. Consistent with this, the fitting procedure selected different exponents for different task types, from 1 through 4, and never selected the same value for all four.

19.4 Radius rules

Finding 19.5 (Radius rules, E3). Under an identical learned geometry, the gap radius attained $F_1 = 0.734$ with a mean width of 5.6 and recovered the relevant set *exactly* on 16.7% of queries. The best fixed- k rule reached $F_1 = 0.678$ at $k = 10$ with exact recovery on 2.7%; the best exact-recovery rate among fixed rules was 7.1% at $k = 3$, where F_1 fell to 0.545. The self-tuning rule attained 0.534 at a mean width of 98.6 and the quantile rule 0.493.

The pattern matches Theorem 7.3 closely. A fixed k can match the relevant set only when k happens to equal its cardinality, so its exact recovery rate is bounded by the probability that $|R| = k$, which for our uniform size distribution over thirteen values is about 7.7% — and $k = 3$ ’s measured 7.1% is essentially that ceiling. The gap rule is

Radius rule	F_1	mean width	exact recovery
R1, rank $k = 3$	0.545	3.0	7.1%
R1, rank $k = 5$	0.657	5.0	3.2%
R1, rank $k = 10$	0.678	10.0	2.7%
R1, rank $k = 20$	0.493	20.0	0.0%
R1, rank $k = 50$	0.255	50.0	0.0%
R2, quantile $q = 0.01$	0.493	20.0	0.0%
R3, gap	0.734	5.6	16.7%
R4, self-tuning	0.534	98.6	7.5%

Table 7: Radius rules under a fixed learned geometry, three seeds.

not bound by it and more than doubles the rate. The self-tuning rule illustrates the opposite failure: it is scale invariant, as Proposition 7.5 promises, but it keys off a single order statistic and therefore inflates the width by a factor of eighteen when the distance spectrum has no sharp elbow at the chosen rank.

19.5 Tool routing

Finding 19.6 (Routing trade-off, E4). With six hundred candidate tools and a selection-accuracy curve $a(k) = (1 + 0.06k)^{-1}$, routing success as a function of the radius rose from 0 at $\varepsilon = 0.05$ to a maximum of 0.645 at $\varepsilon^* = 0.319$, where the correct tool survived routing 87.5% of the time and the exposed set averaged 5.9 tools, and then fell to 0.027 under full exposure.

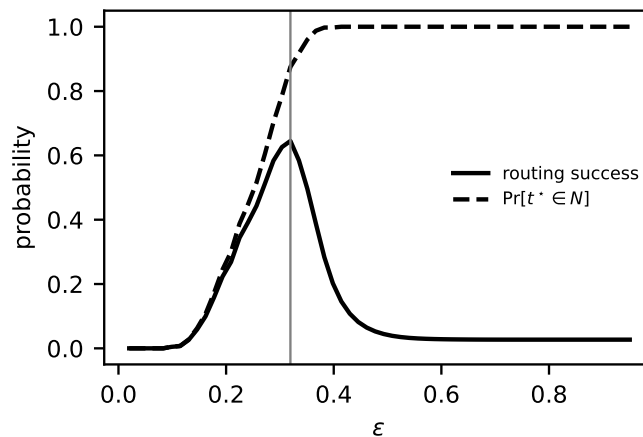


Figure 2: Routing success and survival probability against the radius. The interior maximum predicted by Theorem 11.2 is visible at $\varepsilon^* = 0.319$; the two error sources move in opposite directions on either side of it.

The measured curve is the theorem made visible. Below the optimum the correct tool is often filtered out and recall loss dominates; above it the candidate list grows and selection loss dominates; and the factor of twenty-four between the optimum and full exposure quantifies the cost of declining to select at all. We stress that the accuracy curve is an assumption, not a measurement: its shape is calibrated to the qualitative finding in the tool-use literature that accuracy degrades with list length, and the location of ε^* inherits that assumption even though the existence of an interior optimum does not.

19.6 Planning

Finding 19.7 (Planning bound, E5). On forty random graphs with six hundred states, out-degree twenty-five and transition costs in $[1, 2]$, restricting expansion to five successors chosen by a noisy heuristic gave an empirical covering parameter $\delta = 0.554$ against $c_{\min} = 1$, hence a guaranteed cost ratio of 2.37 by Theorem 13.4. The realised mean cost ratio was 1.003, and the bound held on every trial while the branching factor fell from twenty-five to five.

Two things are worth saying about a bound that holds by a factor of two. The guarantee is a worst case over a supremum: δ is the largest covering failure anywhere in the graph, while realised cost depends on the failures actually encountered along the chosen path, which are typically far smaller. And the practical value of the theorem is not the numeric bound but the fact that δ is measurable: an operator can compute it from logs and know, before deploying a narrower neighbourhood, whether planning quality is at risk.

19.7 Ultrametric caching

Finding 19.8 (Exact context reuse, E6). On a simulated hierarchical workspace of depth four and branching five with an agent making 4000 moves, of which 85% stayed inside the current subtree, an ultrametric structural view yielded exact context reuse on 86.3% of consecutive steps. A flat geometry over the same paths at the same nominal radius reused exactly 18.1% of the time.

This is the largest efficiency effect we measured and it is a direct consequence of Theorem 5.9: under an ultrametric the context is not merely similar after a small state change, it is identical, until the state leaves the block. The practical reading is that a workspace with a genuine hierarchy — a monorepo, a package tree, a document taxonomy — can have its context computed once per block rather than once per step, and that the choice of component distance, not the aggregator or the radius, is what unlocks this.

19.8 Indexing, concentration, drift and stability

Finding 19.9 (Filtering bound, E7). Over four hundred randomised configurations with $N = 4000$ and a radius at the first percentile of the distance spectrum, the Euclidean filter of Theorem 17.2 produced zero false negatives. The candidate set was 32.0% of the database for unconstrained softmax weights, falling to 13.3% when the runtime’s weight floor was applied.

Finding 19.10 (Concentration, E8). Under uniform weights the relative contrast $(\max D - \min D) / \min D$ fell from 370 at $n = 2$ to 1.45 at $n = 16$ and 0.34 at $n = 128$, while the relative standard deviation of the aggregated power sum decayed as $n^{-1/2}$ exactly as Proposition 4.11 predicts. A weighting concentrated on three views held $n_{\text{eff}} = 3$ and contrast above 17 at every n .

Finding 19.10 is the empirical case for the entropy regulariser of Section 16, and it also delimits the framework’s own ambition: adding views is not free, and a system that accumulates thirty weakly informative views under uniform weighting will retrieve worse than one with six, whatever the aggregation rule. The effective dimension, not the view count, is the quantity to watch.

Finding 19.11 (Drift detection, E9). With continuous parameter noise applied at every step and two environment shocks replacing a quarter of the objects, the Jaccard drift

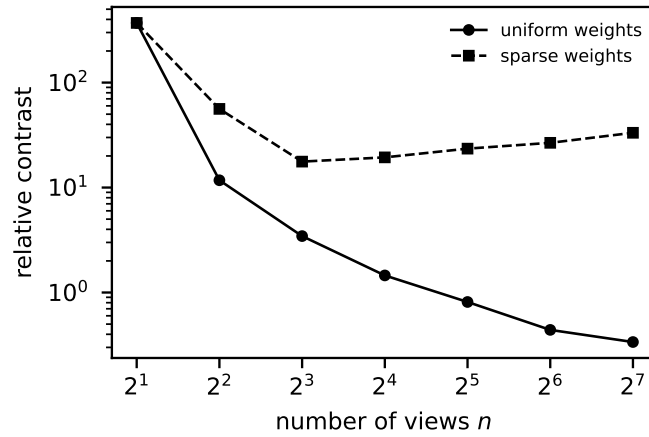


Figure 3: Relative contrast against the number of views, on log–log axes. Uniform weights lose contrast as views accumulate; sparse weights hold n_{eff} and therefore contrast fixed.

statistic had quiet-period mean 0.093 and standard deviation 0.063; a three-sigma threshold at 0.283 caught both shocks, whose peaks reached 0.541 and 0.476, with a false-alarm rate of 0.7% per step.

Finding 19.12 (Stability and constraints, E10). Over three hundred randomised trials with weight perturbations of $\|w - w'\|_1 = 0.05$, the bound of Theorem 6.5 held on every trial; the symmetric difference between the perturbed and unperturbed contexts averaged 11.3% of the context and was contained in the annulus predicted by Theorem 6.7 on every trial; and with $p = \infty$ and $\varepsilon \leq 1$ no policy-violating object entered any context, as Theorem 11.1 requires.

Claim	Prediction	Measurement
Gradients (Prop. 16.3)	exact	median relative error 2×10^{-10}
Rank crossover (Prop. 4.7)	$p^* = 1.3569154$	1.356925 numerically
Exact recovery (Thm. 7.3)	adaptive $>$ any fixed k	16.7% against $\leq 7.1\%$
Routing optimum (Thm. 11.2)	interior maximum	$\varepsilon^* = 0.319, 0.645$ against 0.027
Planning bound (Thm. 13.4)	ratio ≤ 2.37	realised 1.003; held on all trials
Exact reuse (Thm. 5.9)	identical context inside a block	86.3% against 18.1%
Filtering (Thm. 17.2)	no false negatives	0 of 400 trials
Concentration (Prop. 4.11)	contrast decays in n_{eff}	$370 \rightarrow 0.34$ uniform; flat when sparse
Weight stability (Thm. 6.5)	$ \Delta D \leq \ \Delta w\ _1^{1/p}$	held on all 300 trials
Annulus containment (Thm. 6.7)	symmetric difference in A_τ	held on all 300 trials
Constraints (Thm. 11.1)	no violator admitted	0 violations

Table 8: Every theoretical claim we were able to test operationally, with its prediction and its measurement.

20 What the Simulation Does and Does Not Show

The generator embodies the hypothesis under test. Our environment is built so that relevance depends on a task-dependent subset of views and so that semantic proximity is informative but misleading in controlled ways. If real agent workloads have that structure then the measured effects estimate what is available; if they do not, the measurements say only that the machinery works when the hypothesis holds. This is the central limitation and no amount of additional synthetic experimentation removes it. Finding 19.3 is our partial mitigation: by sweeping the parameter that controls the strength of the assumption, we show how the comparison behaves as it weakens, and at zero decoys the advantage shrinks from a factor of 2.8 to a factor of 1.7.

The oracle is an upper bound only within the generator. The oracle knows the generator’s active views. It is a ceiling for the fitting procedure, not a ceiling for the framework, since a richer view system might exceed it on real data — or might not exist at all.

The selection-accuracy curve in E4 is assumed. Its qualitative shape is supported by the tool-use literature, but its parameters are ours. The existence of an interior optimum follows from Theorem 11.2 for any non-increasing accuracy curve; its location does not, and the reported ε^* should be read as an illustration rather than a recommendation.

The bounds we validated cannot fail, and that is the point. Findings 19.7, 19.9 and 19.12 test theorems. A violation would have indicated an implementation error, not a false theorem, and this is exactly why we ran them: the value of the checks is that they falsify the *code*, and E0’s outlier shows they are capable of finding something.

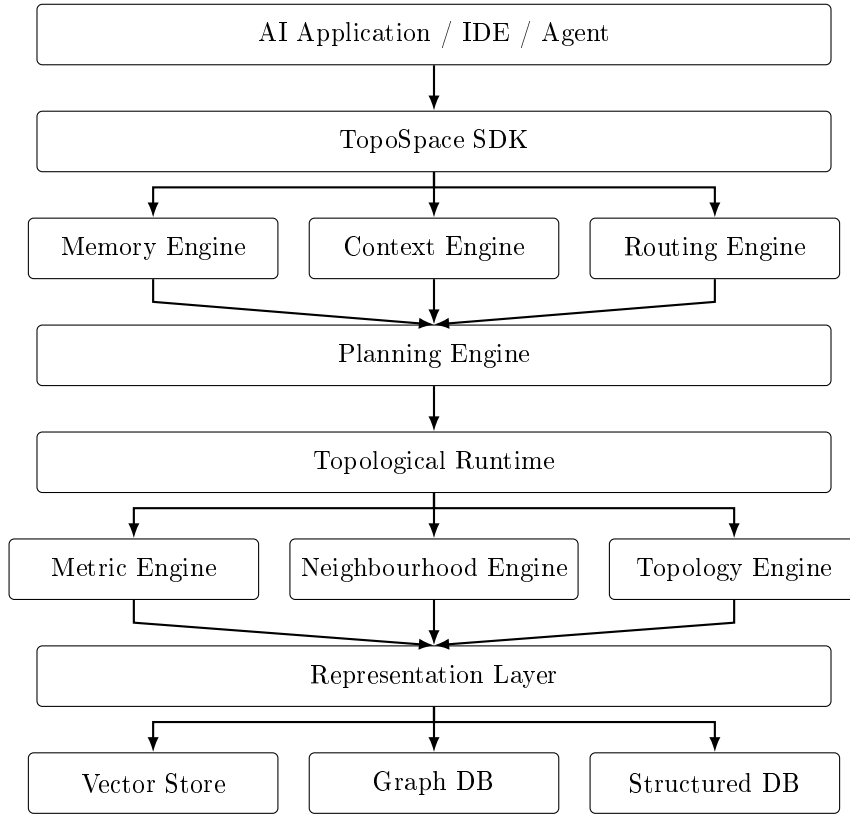
Learning was task-type-conditioned, not state-conditioned. Our adaptive condition fits one geometry per task type, which is the coarsest form of adaptivity. A state-conditioned encoder as in Definition 16.1 is the intended production form, and it is both more powerful and more prone to overfitting; we did not evaluate it, and the gap between task-conditioned and state-conditioned adaptivity is open.

No language model was involved. The study measures context construction, not end-to-end task success. The link between context quality and task success is mediated by the model and is exactly what the field protocol of Section 25 is designed to measure.

Part VI — Systems, Evaluation, and Positioning

21 System Architecture

TopoSpace is infrastructure rather than a model. It sits above heterogeneous storage and below the reasoning layer, and it is designed so that an existing stack can adopt it without replacing anything.



Representation layer. Adapters over existing stores. A vector store supplies the semantic view, a graph database the structural view, a relational store the numeric views. Nothing is migrated, which is what makes incremental adoption possible.

Metric engine. Implements Definition 4.1 and the gradients of Proposition 16.3, validates that admitted views satisfy the hypotheses of Theorem 4.3, and records the class of each view so that the applicable guarantees of Table 4 can be reported rather than assumed.

Neighbourhood engine. Radius rules R1 through R5, the filtration of Section 8, and the candidate-generation bound of Theorem 17.2 including the weight floor whose effect Finding 19.9 quantifies.

Topology engine. Vietoris–Rips construction, persistence, and the drift statistics of Definition 15.2, with alarm thresholds derived from Proposition 15.5 rather than hand-tuned.

Planning engine. Restricted search per Section 13, exporting the empirical covering parameter δ as an operational metric, since Theorem 13.4 makes it the quantity that predicts plan degradation.

Context, memory and routing engines, and the SDK. Task-facing compositions of the layers beneath, corresponding to Sections 10 through 12, exposed through the developer surface of Section 23.

22 Open-Source Architecture

```

topospace/
|
+-- core/
|   +-- metric/           # weighted  $L^p$ , aggregators, view registry, validation
|   +-- topology/         # Vietoris-Rips, persistence, drift
|   +-- neighborhood/     # radius rules R1-R5, filtrations
|   +-- state/            # state and goal encoders
|   +-- learning/         # gradients, relaxation, two-stage fit
|
+-- sdk/
|   +-- python/
|   +-- typescript/
|
+-- runtime/              # local and remote execution, caching
+-- mcp/                  # MCP server exposing the runtime
+-- skills/
|   +-- context/ memory/ tool-router/ planner/ multi-agent/
|
+-- integrations/         # vector stores, graph DBs, agent frameworks
+-- sim/                 # the simulator of Part V, reproducing every number
+-- benchmarks/          # the field protocol of Section 25
+-- examples/            # coding-agent reference application

```

The core engine defines the mathematical primitives and is the artefact against which the theorems of Part II can be checked. The SDK exposes developer abstractions, the runtime provides persistent or remote execution, and the skills are clients of the infrastructure rather than the infrastructure itself, which preserves independence from any single agent ecosystem. The simulator is shipped alongside rather than kept internal, because a performance claim whose harness is unavailable is not a claim a reader can evaluate.

Open core. The open distribution comprises the mathematical core, the L^p and weighted metrics together with the admissible aggregators, adaptive neighbourhoods, local state-space construction, both SDKs, the local runtime, the MCP interface, the context, memory and tool-routing skills, the simulator, the benchmark suite and the coding-agent examples. Commercial extensions are orthogonal to the mathematics: distributed state spaces, high-performance and accelerated indexing, persistent enterprise state, large-scale orchestration, observability, enterprise security and managed hosting.

23 Developer Interface

Mathematical complexity should remain internal, so the default surface is small.

```

from topospace import Space

space = Space() # defaults: p=2, uniform weights, gap radius

space.add(memory); space.add(tool); space.add(skill); space.add(agent)

context = space.context(current_state) # the neighbourhood
memories = space.recall(current_state) # neighbourhood intersect M
tools = space.route(current_state, type="tool")
agents = space.route(current_state, type="agent")

```

Advanced users configure the geometry explicitly, including the grouped aggregation licensed by Proposition 5.3.

```
space = Space(
    aggregator={"groups": {"evidence": 1.0, "policy": "inf"}, "across": 2.0},
    p="learned", # 1 + softplus(...) (Definition 16.2)
    epsilon="gap", # radius rule R3
    dimensions=["semantic", "structural", "temporal",
                "causal", "cost", "risk", "security"],
    constraints={"security": "hard"}, # forces the bottleneck geometry
    hierarchy={"module_path": "ultrametric"} # enables exact block caching
)

tiers = space.filtration(current_state, scales=[0.2, 0.5, 0.9])
report = space.drift(window=50)
audit = space.guarantees() # which theorems hold for this configuration
```

Three arguments deserve comment. Declaring a view `hard` is not a filter bolted on after ranking but a request for the non-compensatory geometry of Theorem 11.1, whose exclusion guarantee then follows from the distance function and can be asserted in a unit test. Declaring a field `ultrametric` activates Theorem 5.9 and with it the block-level cache whose effect Finding 19.8 measures. And `guarantees()` returns the row of Table 4 that the current configuration actually earns, which is the feature that keeps the formal apparatus connected to the running system.

24 MCP and Skill Infrastructure

The runtime is exposed through an MCP-compatible interface so that any MCP-speaking agent can use it without linking the SDK.

<code>topospace.context</code>	-> neighbourhood of the current state
<code>topospace.recall</code>	-> memory neighbourhood
<code>topospace.route_tools</code>	-> tool neighbourhood
<code>topospace.route_agents</code>	-> agent neighbourhood
<code>topospace.plan</code>	-> restricted planning (Section 13)
<code>topospace.update_state</code>	-> state transition, drift bookkeeping
<code>topospace.guarantees</code>	-> applicable guarantees for the configuration

Reference skills for context, memory, tool routing, planning and multi-agent routing are distributed as thin clients of these endpoints. Keeping the skills on the client side of the boundary is deliberate: the infrastructure then has no dependency on any particular agent framework's skill format, and a change in that format costs a client rewrite rather than a core rewrite.

25 Field Benchmark Protocol

Part V measured mechanisms in a world we built. This section states what we will measure in a world we did not, before the fact, so that the results are falsifiable.

25.1 Baselines and metrics

The baselines are those of Section 18 instantiated on real corpora: embedding Top- k for several k , hybrid sparse-plus-dense retrieval, m -hop graph retrieval, full-context exposure and hand-written rule-based routing. The context metrics are recall, precision and token reduction against the objects the agent actually used; the routing metrics are tool recall at k , selection accuracy and routing latency; the coding-agent metrics are task success rate, files read, context tokens, agent steps and test pass rate; the multi-agent metrics are agent selection accuracy, execution cost, completion rate and coordination steps.

25.2 Hypotheses and falsification conditions

H1, efficiency. At matched task success rate, TopoSpace consumes strictly fewer context tokens than the best baseline. Refuted if no configuration matches baseline success at lower token cost.

H2, precision at matched recall. At matched context recall, TopoSpace achieves higher context precision. Refuted if the precision–recall curves coincide within noise or cross unfavourably.

H3, adaptivity pays. Learned state-dependent geometry outperforms the best fixed geometry found by grid search on the same data. Refuted if the fixed optimum is within noise of the adaptive one. Our simulation supports H3 strongly, at 0.705 against 0.402, but under a generator built to reward adaptivity; this is the hypothesis whose field replication we consider least certain and most important.

H4, drift utility. Drift-triggered context invalidation improves success or reduces cost relative to fixed-interval invalidation. Refuted if the trigger fires no more informatively than a timer.

H5, ultrametric caching. Adding an ultrametric structural view raises the context cache hit rate at matched task success. Refuted if the hit rate does not improve, or if success drops once the ultrametric view is weighted heavily enough to matter. Our simulation measured 86.3% against 18.1% exact reuse, which sets the expectation this hypothesis must meet in the field.

H6, component distances. For at least one view, replacing the default component distance with an alternative from Appendix B — Jensen–Shannon or Wasserstein for outcome profiles, resistance or diffusion for the dependency graph, tree edit distance for code — improves precision at matched recall. Refuted if the angular-plus-hop default is never beaten.

26 Limitations

No field validation. Part V is a simulation. Every claim about performance on real agent workloads remains a hypothesis, and Section 20 lists the specific ways the simulation could mislead.

View engineering is the real work. Theorem 4.3 guarantees that good views compose into a good metric; it says nothing about where good views come from. A poorly calibrated causal view produces a confidently wrong geometry. The framework relocates difficulty from retrieval tuning to view design, which is an improvement only if view design is more tractable — itself an empirical question, though the catalogue of Appendix B is our attempt to make it a matter of selection rather than invention.

The topology is thin where the name is loudest. Remark 4.9 is not a footnote. On a finite object set the metric topology carries no information, and the non-trivial topological content is confined to filtrations and drift. Readers should discount any impression that homological machinery is doing work in the core retrieval path; it is not.

Adaptivity risks instability and overfitting. A geometry that changes every step can produce context churn. Theorem 6.7 bounds this and Finding 19.12 measures it at 11.3% under a specific perturbation, but only in terms of quantities that must themselves be controlled by regularisation and conservative learning rates. Our study also fit a coarse, task-conditioned adaptivity; the state-conditioned form is more expressive and more exposed.

Asymmetry limits composition. Proposition 6.2 means that filtration, persistence and the delegation bound require symmetry or a fixed geometry over the relevant scope. Implementations that vary geometry inside a planning horizon fall outside Theorems 13.3 and 13.4.

Diversity is outside the axioms. Theorem 9.5 assumes threshold consistency, which diversity-aware selection deliberately violates. Treating diversity as post-processing is a factorisation, not a proof that the factorisation is right.

Indexing costs are real. Finding 19.9 measured a candidate set of 13% to 32% of the database. The bound is tight enough to be useful at moderate scale and loose enough to matter at very large scale, and improving it — perhaps by maintaining a small number of indexes at different exponents — is open.

Operational overhead. Section 17 argues the computational overhead is small beside a model call, but another service, another set of parameters and another failure mode are real costs that small deployments will not amortise.

27 Research Directions

Learned metric validity beyond the sufficient conditions. Proposition 16.2 gives sufficient conditions for validity. What is the largest useful class of learned dissimilarities for which Part II survives in approximate form, for instance under an ϵ -relaxed triangle inequality?

Variable exponents. What structure arises when $p = p(x)$ or $p = p(x, y)$ rather than $p = p(s)$? Variable-exponent Lebesgue spaces have a developed theory [10]; which parts transfer to this finite, weighted, learned setting?

Sharper stability. Corollary 8.5 bounds diagram movement under parameter change. Are there dimension-free improvements exploiting the special structure of aggregation over few views?

End-to-end radius learning. Can ε be learned from downstream performance, and does the learned rule converge to gap-like behaviour, as Theorem 7.3 and Finding 19.5 together suggest it should?

Multi-scale importance. Do objects with long persistence across the filtration predict contextual importance better than rank at any single scale?

Topological planning. Can invariants of the context — component counts, cycle structure — reduce long-horizon search complexity beyond the branching reduction of Theorem 13.3?

Concentration-aware geometry. Proposition 4.11 and Finding 19.10 tie contrast to the effective dimension. Is there an optimal joint schedule for the exponent and the weights as views accumulate?

From task-conditioned to state-conditioned adaptivity. Our study fit one geometry per task type. Characterising when the finer, state-conditioned parameterisation of Definition 16.1 helps and when it overfits is the most immediate open question raised by the results.

28 Positioning

A vector database answers *what is similar?* A graph system answers *what is connected?* TopoSpace answers

what belongs to the relevant neighbourhood of the current state?

which for an autonomous agent is

what should the agent consider next?

The proposed category is **AI State and Context Infrastructure** and the underlying technology is the **adaptive topological state space**. The distinction from a vector database is not a marketing one: a vector database is a component of the representation layer of Section 21, and TopoSpace consumes it.

29 Vision

The current stack can be summarised as Model + Retrieval + Tools + Agent. TopoSpace proposes an additional layer,

Model + State Space + Context + Tools + Agent

in which the model provides reasoning, storage provides information, tools provide actions, and TopoSpace provides the space in which relevance is determined. The long-term objective is a general mathematical infrastructure in which memories, knowledge, tools,

skills, agents, code, actions and objectives coexist in dynamically constructed state spaces, so that instead of inspecting everything it can address, an agent continuously answers a more fundamental question:

Given where I am now and what I am trying to accomplish, what part of my world should matter next?

A Deferred Proofs

A.1 Minkowski's inequality, weighted form

Proof of Lemma 4.2. For $p = 1$ the claim is the triangle inequality for real numbers summed against non-negative weights. Let $p > 1$ and let $q = p/(p - 1)$ be the conjugate exponent. Write μ for the measure on $\{1, \dots, n\}$ with $\mu(\{i\}) = w_i$ and $\|f\|_p = (\sum_i w_i |f_i|^p)^{1/p}$. Hölder's inequality in this measure states $\sum_i w_i |f_i g_i| \leq \|f\|_p \|g\|_q$; it follows from Young's inequality $ab \leq a^p/p + b^q/q$ applied to the normalised functions $f/\|f\|_p$ and $g/\|g\|_q$ and summing.

Now, with $a, b \geq 0$ and $S = \|a + b\|_p$, assume $S > 0$ (else there is nothing to prove). Then

$$S^p = \sum_i w_i (a_i + b_i)^p = \sum_i w_i a_i (a_i + b_i)^{p-1} + \sum_i w_i b_i (a_i + b_i)^{p-1}.$$

Applying Hölder to each term with the pair (p, q) and using $(p - 1)q = p$,

$$S^p \leq (\|a\|_p + \|b\|_p) \left(\sum_i w_i (a_i + b_i)^p \right)^{1/q} = (\|a\|_p + \|b\|_p) S^{p/q}.$$

Dividing by $S^{p/q} > 0$ and using $p - p/q = 1$ gives $S \leq \|a\|_p + \|b\|_p$, as claimed. $\square$

A.2 Norm comparison

Proof of Lemma 17.1. For the left inequality assume $\|v\|_p = 1$ (the general case follows by homogeneity). Then $|v_i| \leq 1$ for all i , so $|v_i|^q \leq |v_i|^p$ for $q \geq p$, whence $\|v\|_q^q \leq \|v\|_p^p = 1$ and $\|v\|_q \leq 1 = \|v\|_p$.

For the right inequality apply Hölder's inequality to the pair $(|v_i|^p, 1)$ with exponents $q/p > 1$ and $r = (1 - p/q)^{-1}$:

$$\sum_i |v_i|^p \cdot 1 \leq \left(\sum_i |v_i|^q \right)^{p/q} \left(\sum_i 1^r \right)^{1/r} = \|v\|_q^p n^{1-p/q}.$$

Raising to the power $1/p$ gives $\|v\|_p \leq n^{\frac{1}{p} - \frac{1}{q}} \|v\|_q$. The case $q = \infty$ follows by taking limits. $\square$

B Catalogue of Admissible View Distances

This appendix lists distances that can serve as component views δ_i in Definition 3.2, grouped by the kind of object they compare. The theory governing their use is Section 5: Theorem 5.2 says any collection of pseudometrics may be combined by any admissible aggregator, and Table 4 says which results survive when a component is not a metric.

Class codes. M metric PM pseudometric (may vanish off the diagonal) UM ultra-metric (strong triangle inequality; Theorem 5.9) QM quasi-metric (asymmetric) D divergence (asymmetric, no triangle inequality) NM symmetric but not a metric (triangle inequality fails).

B.1 How to choose a component distance

The tables that follow are long, but the selection procedure is short, and it is worth stating explicitly because most of the errors we have seen in practice are made at step three rather than step one.

Object type. Begin from what the view actually compares — a vector, a distribution, a set, a string, a graph, a hierarchy, a timestamp, a category — rather than from what is convenient to compute. The tables are organised by this axis precisely because it is the one that determines correctness; computational convenience determines only which row within a table is affordable.

Metric class. Prefer an entry marked M, PM or UM, and among those prefer bounded ranges. Boundedness is not cosmetic: cross-view weights are only interpretable when the views they weight live on comparable scales, and an unbounded view will silently dominate the aggregate whenever a single outlier appears. This is the reason the reference configurations of Section B.10 favour angular over raw Euclidean distance and exponential decay over raw lag.

Metrification. If the natural comparison is a divergence (D) or a non-metric dissimilarity (NM), apply one of the recipes (M1)–(M4) of Section 5. Most practitioners reach for KL because it is familiar; the Jensen–Shannon square root is available, is bounded, and is an actual metric, so the familiar choice is almost never the right one.

Raw admission. If no recipe applies, admit the view as raw and consult Table 4 to see what has been given up. Ranking, radius selection, exact recovery and constraint enforcement survive; metric deduplication, delegation containment, persistence stability and drift as a pseudometric do not. A raw view is therefore acceptable in a retrieval-only deployment and dangerous in a multi-agent one.

Normalisation and registration. Compose with a concave, bounded ψ (Lemma 3.3) to bring the view into $[0, 1]$; this preserves the metric class, so nothing is lost. Finally, register the class with the runtime, which is what allows the system to report which of the guarantees in Table 4 currently hold for the deployed configuration rather than leaving the operator to work it out from the paper.

B.2 A. Vectors and embeddings

Distance	Form	Class	Range	TopoSpace use
Weighted Minkowski L^p	$(\sum_i w_i \Delta_i ^p)^{1/p}, p \geq 1$	M/PM	$[0, \infty)$	default aggregator (Def. 4.1)
Euclidean L^2	$\ \Delta\ _2$	M	$[0, \infty)$	numeric coordinates
Manhattan L^1	$\sum_i \Delta_i $	M	$[0, \infty)$	compensatory views
Chebyshev L^∞	$\max_i \Delta_i $	M	$[0, \infty)$	bottleneck / constraints
Weighted Chebyshev	$\max_i w_i \Delta_i $	M	$[0, \infty)$	graded bottleneck (Rem. 5.6)
Mahalanobis	$\sqrt{\Delta^\top A \Delta}, A \succeq 0$	M/PM	$[0, \infty)$	correlated numeric views
Angular distance	$\frac{2}{\pi} \arccos \cos(x, y)$	M	$[0, 1]$	semantic view (default)
Chord distance	$\sqrt{2 - 2 \cos(x, y)}$	M	$[0, 2]$	semantic view, index-friendly
Correlation distance	$\sqrt{2(1 - \rho)}$	M	$[0, 2]$	profile / behaviour vectors
Cosine dissimilarity	$1 - \cos(x, y)$	NM	$[0, 2]$	<i>avoid</i> ; use angular instead
Canberra	$\sum_i \frac{ x_i - y_i }{ x_i + y_i }$	M	$[0, n]$	counts, sparse usage vectors
χ^2 distance	weighted Euclidean of profiles	M	$[0, \infty)$	frequency profiles
Energy distance (root)	$\sqrt{\mathcal{E}(X, Y)}$	M	$[0, \infty)$	comparing sample sets
Bray–Curtis	$\frac{\sum x_i - y_i }{\sum (x_i + y_i)}$	NM	$[0, 1]$	raw view only
Fractional $L^p, p < 1$	$(\sum w_i \Delta_i ^p)^{1/p}$	NM	$[0, \infty)$	excluded (Prop. 4.5)
Power sum, $p \leq 1$	$\sum_i w_i \Delta_i ^p$	M	$[0, \infty)$	admissible alternative to the above

B.3 B. Probability distributions and uncertainty

Used where an object is characterised by a distribution: a tool’s outcome or latency profile, an agent’s competence profile, a retrieval score distribution, a model’s predictive uncertainty.

Distance	Form	Class	Range	TopoSpace use
Total variation	$\frac{1}{2} \sum_i p_i - q_i $	M	$[0, 1]$	outcome profiles
Hellinger	$\frac{1}{\sqrt{2}} \ \sqrt{p} - \sqrt{q}\ _2$	M	$[0, 1]$	bounded, index-friendly
Jensen–Shannon distance	$\sqrt{\text{JSD}(p\ q)}$	M	$[0, 1]$	default distributional view [14]
Wasserstein- q	$\inf_\pi (\int \int c^q d\pi)^{1/q}$	M	$[0, \infty)$	latency/cost distributions [33]
Sliced Wasserstein	averaged 1-D W_q over projections	M	$[0, \infty)$	cheap W surrogate [27]
Maximum mean discrepancy	$\ \mu_p - \mu_q\ _{\mathcal{H}}$	PM/M	$[0, \infty)$	sample sets; M iff kernel characteristic [15]
Fisher–Rao	$2 \arccos \sum_i \sqrt{p_i q_i}$	M	$[0, \pi]$	information- geometric view
Kullback–Leibler	$\sum_i p_i \log(p_i/q_i)$	D	$[0, \infty]$	raw view; metrise via JS
Jeffreys (symmetrised KL)	$\text{KL}(p\ q) + \text{KL}(q\ p)$	NM	$[0, \infty]$	raw view
Bregman divergence	$F(p) - F(q) - \langle \nabla F(q), p - q \rangle$	D	$[0, \infty)$	raw view; ranking only

B.4 C. Sets, strings, sequences, and rankings

Distance	Form	Class	Range	TopoSpace use
Jaccard distance	$1 - \frac{ A \cap B }{ A \cup B }$	M	$[0, 1]$	tags, permissions, task sets
Symmetric difference	$ A \triangle B $	M	$[0, \infty)$	drift bookkeeping (Thm. 15.3)
Sørensen–Dice dissimilarity	$1 - \frac{2 A \cap B }{ A + B }$	NM	$[0, 1]$	raw view
Hamming	$\#\{i : x_i \neq y_i\}$	M	$[0, n]$	flags, feature masks, SimHash
Levenshtein	minimum edit cost	M	$[0, \infty)$	identifiers, short strings
LCS distance	$ x + y - 2 \text{lcs}$	M	$[0, \infty)$	diffs, log lines
q -gram / token Jaccard	Jaccard on q -gram sets	M	$[0, 1]$	code tokens, cheap surrogate
Normalised compression distance	$\frac{C(xy) - \min(C(x), C(y))}{\max(C(x), C(y))}$	NM*	$[0, 1]$	model-free file similarity [5]
Kendall tau distance	$\#$ discordant pairs	M	$[0, \binom{n}{2}]$	comparing rankings/plans
Spearman footrule	$\sum_i \sigma(i) - \tau(i) $	M	$[0, \infty)$	context-stability metric
Dynamic time warping	optimal warping cost	NM	$[0, \infty)$	raw view; trajectories
Fréchet distance	curve-matching distance	M	$[0, \infty)$	trajectory comparison

* satisfies the metric axioms only up to an additive $O(\log)$ error; treat as raw unless the error is negligible.

B.5 D. Code and graph structure

Distance	Form	Class	Range	TopoSpace use
Shortest path (geodesic)	hop or weighted path length	M	$[0, \infty]$	structural view (default)
Capped hop distance	$\psi(\text{hops})$, ψ concave	M	$[0, 1]$	bounded structural view
Resistance distance	$\Omega_{xy} = (L^+)_{xx} + (L^+)_{yy} - 2(L^+)_{xy}$	M	$[0, \infty)$	robust to single-edge noise [18]
Commute-time distance	$\propto$ resistance distance	M	$[0, \infty)$	random-walk proximity
Diffusion distance at t	$\ P^t(x, \cdot) - P^t(y, \cdot)\ _{1/\pi}$	M	$[0, \infty)$	multi-scale structure [7]
Tree edit distance	minimum AST edit cost	M	$[0, \infty)$	code / config view
Graph edit distance	minimum graph edit cost	M [†]	$[0, \infty)$	schema, workflow comparison
WL kernel distance	$\sqrt{2 - 2\kappa_{\text{WL}}}$	PM	$[0, 2]$	subgraph-structure view [30]
Personalised PageRank	$1 - \text{ppr}(x \rightarrow y)$	QM	$[0, 1]$	directional relevance (causal view)
SimRank	$1 - \text{simrank}(x, y)$	NM	$[0, 1]$	raw view

[†] a metric provided the elementary edit costs themselves form a metric.

B.6 E. Hierarchies, versions, and lineage (ultrametric family)

These are the views that activate Theorem 5.9 and its exact cache-reuse consequence.

Distance	Form	Class	Range	TopoSpace use
Longest-common-prefix	$2^{-\text{lcp}(x, y)}$	UM	$[0, 1]$	file paths, module names
p -adic absolute value	$ x - y _p = p^{-v_p(x - y)}$	UM	$[0, 1]$	versioned / lineage identifiers [19]
Cophenetic (dendrogram)	merge height of the lowest common ancestor	UM	$[0, 1]$	hierarchical clustering of memories
Taxonomy path distance	weighted path through the LCA	M	$[0, 1]$	tag ontologies, skill taxonomies
Tree metric	path length in a weighted tree	M	$[0, \infty)$	org charts, package trees
Hyperbolic (Poincaré)	$\text{arcosh}\left(1 + 2\frac{\ x - y\ ^2}{(1 - \ x\ ^2)(1 - \ y\ ^2)}\right)$	M	$[0, \infty)$	low-distortion hierarchy embedding [25]
Version-precedence	asymmetric ancestry depth	QM	$[0, 1]$	“supersedes” relations

B.7 F. Time and schedule

Distance	Form	Class	Range	TopoSpace use
Absolute lag	$ t(x) - t(y) $	M	$[0, \infty)$	raw recency
Log lag	$\log(1 + \Delta t)$	M	$[0, \infty)$	long-horizon memories
Exponential decay	$1 - e^{-\lambda \Delta t }$	M	$[0, 1)$	temporal view (default)
Saturating lag	$ \Delta t / (1 + \Delta t)$	M	$[0, 1)$	bounded alternative
Circular distance	$\min(\Delta , T - \Delta)$	M	$[0, T/2]$	periodic schedules, on-call cycles
Interval (Allen) relations	relation-graph distance	NM	—	raw view

B.8 G. Categorical, policy, and security views

Distance	Form	Class	Range	TopoSpace use
Discrete metric	$\mathbf{1}[x \neq y]$	M	$\{0, 1\}$	tenant, region, principal
Constraint indicator	$\mathbf{1}[x \text{ violates policy at } s]$	M	$\{0, 1\}$	hard constraints (Thm. 11.1)
Weighted Hamming	$\sum_i w_i \mathbf{1}[x_i \neq y_i]$	M	$[0, \sum w_i]$	attribute profiles
Lattice distance	graded path length in the permission lattice	$M^\dagger$	$[0, \infty)$	role hierarchies
Clearance gap	$\max(0, \text{req}(x) - \text{lvl}(s))$	QM	$[0, \infty)$	directional authorisation

† a metric on graded lattices, where the rank function induces a well-defined path length.

B.9 H. Environment-level (cross-space) distances

These do not compare objects but whole environments or contexts, and are used by the drift machinery of Section 15 and the evaluation protocol of Section 25.

Distance	Form	Class	Range	TopoSpace use
Neighbourhood Jaccard	Steinhaus transform of $ \Delta $	M	$[0, 1]$	per-step drift (default)
Sup-norm on distances	$\sup_{x,y} d - d' $	PM	$[0, \infty)$	geometry drift
Hausdorff	$\max(\sup_a \inf_b d, \sup_b \inf_a d)$	M	$[0, \infty)$	context-set movement
Bottleneck on diagrams	$d_B(\text{Dgm}, \text{Dgm}')$	M	$[0, \infty)$	topological drift (Thm. 8.4)
Wasserstein on diagrams	$W_q(\text{Dgm}, \text{Dgm}')$	M	$[0, \infty)$	noise-tolerant variant
Gromov–Hausdorff	inf over correspondences of distortion	M	$[0, \infty)$	environments with different object sets
Gromov–Wasserstein	Prop. 5.10	M	$[0, \infty)$	intrinsic environment drift [24]
Kendall tau on contexts	rank correlation of the context ordering	M	$[0, 1]$	stability under parameter updates

B.10 Reference configurations

The catalogue is a menu, not a recommendation. The three configurations below are the ones we would defend as starting points, each valid by Theorem 5.2 and each corresponding to a deployment shape that appears repeatedly in practice. All three were exercised, in reduced form, by the simulator of Appendix C.

Retrieval-first. Views: angular (semantic), exponential decay (temporal), Jaccard on task tags. Aggregator: L^1 . Radius: R3, the gap rule. This is the drop-in replacement for a conventional retrieval stack, and it degenerates exactly to Top- k semantic search when the non-semantic weights are set to zero (Proposition 4.10), which makes it the configuration to deploy first: it cannot underperform the incumbent by construction, and every subsequent weight the learner moves away from zero is an auditable claim that a non-semantic signal carries information. In the simulation this configuration corresponds to the gap between the `sem` and `ts-adapt` rows of Table 6.

Coding agent. Views: angular (semantic), capped hop distance on the import graph (structural), personalised PageRank from the stack trace (causal, QM), exponential decay (temporal), longest-common-prefix on module paths (UM), constraint indicator (security). Aggregator: nested — L^1 within the semantic and temporal group, L^∞ over the security group, L^2 across groups, which is admissible by Proposition 5.3(iv). Radius: R3, with the ultrametric block cache of Theorem 5.9. The ultrametric view is what makes this configuration cheap rather than merely accurate: the measured reuse rate of Finding 19.8 is the reason to include it even when its weight is small.

Tool routing under policy. Views: angular (capability), Jensen–Shannon (outcome profile), Wasserstein-1 (latency profile), Jaccard (input/output type match), constraint indicator (security). Aggregator: weighted Chebyshev. Radius: R4, self-tuning. The Chebyshev aggregator is the load-bearing choice here, since it makes Theorem 11.1 apply for every $\varepsilon \leq 1$: no exposed tool can violate a policy regardless of how attractive it is on the remaining views. Finding 19.6 quantifies what this buys against full tool exposure.

C Reference Simulator: Full Specification

This appendix specifies the simulator used in Part V in enough detail that the numbers reported there can be reproduced or contested without reading the source. The implementation accompanies the paper as two files, `topospace_core.py` (geometry, radius rules, generator, learner) and `experiments.py` (the eleven experiments and the figures); the whole suite runs in 47 seconds on a single core and depends only on NumPy.

C.1 Why a synthetic environment

A synthetic generator is the wrong instrument for estimating how TopoSpace would perform on a real codebase, and we make no such claim. It is the right instrument for a different and prior question: whether the mathematics in Parts II–IV does what the proofs say it does, and whether the central hypothesis is even coherent. A theorem about neighbourhood stability can be checked against measured symmetric differences; a claimed crossover exponent can be checked against a numerically located one; a gradient derived by hand can be checked against finite differences. None of that requires realism, and all of

it is a precondition for deployment. The generator’s role, in short, is to make the theory falsifiable at low cost, and Section 20 states plainly what it cannot establish.

C.2 Ground truth: how relevance is decided

The generator embodies the multi-view hypothesis directly. Each state carries $N = 2000$ candidate objects described by six view values in $[0, 1]$ — semantic, structural, causal, temporal, cost, security — and a hidden task type drawn from four possibilities. Each task type fixes an *active subset* of views and a true exponent; relevance is decided by the active views alone, so that any method restricted to a single view is structurally unable to recover the relevant set. The four types use the active sets $\{\text{sem}, \text{cau}\}$, $\{\text{str}, \text{cau}, \text{tim}\}$, $\{\text{sem}, \text{str}\}$ and $\{\text{cau}, \text{tim}\}$ with exponents 1, 2, 4, 8 respectively, which exercises the full range of compensatory to bottleneck behaviour discussed in Section 4.

Backgrounds. View values for the general population are drawn from $\text{Beta}(5, 5)$, concentrated near 0.5. This detail matters more than it appears to. An earlier calibration used uniform backgrounds, under which being semantically near is unremarkable and every semantic baseline collapses to noise; the resulting comparison flattered TopoSpace for the wrong reason. A concentrated background reproduces the property that makes retrieval work in practice — proximity is rare, and therefore informative — and it is the reason the semantic baselines in Table 6 are non-trivial competitors rather than straw men.

Positives. A relevant set of size $|R| \sim \mathcal{U}\{3, \dots, 15\}$ is drawn; on each active view its members take values from $\text{Beta}(1.5, 10)$, i.e. genuinely near. The semantic view is made informative for *every* task type, whether or not it is active, so that semantic similarity is a real signal and not a decoy: this is the assumption most favourable to the baselines.

Decoys and remote positives. Two perturbations break the sufficiency of that signal. First, $\text{decoy_mult} \times |R|$ irrelevant objects (default multiplier 2) are given semantic values drawn from the positive distribution, so they are near in text and wrong in substance. Second, 30% of the positives are pushed to semantic values in $\mathcal{U}(0.55, 1)$, so they are relevant and textually remote. These are the two failure modes that motivate the entire paper, and Finding 19.3 reports the sensitivity sweep over the decoy multiplier, which is the honest way to present a knob of this kind: the advantage narrows as decoys are removed, and the sweep says by how much.

Margin enforcement. After decoy injection, every negative is required to violate at least one active view by a margin of 0.35; violations are repaired by resampling until the condition holds. Enforcing the margin *after* rather than before decoy injection is the difference between a generator with a well-defined separation guarantee and one whose guarantee is silently destroyed by the decoys, and it is what makes the exact-recovery rate in Table 7 interpretable in the light of Theorem 7.3.

Security. Five per cent of negatives carry a security view value of 1 and are never relevant, which is what allows Finding 19.6 to measure constraint violations rather than merely assert Theorem 11.1.

Tokens. Each object is assigned a token count drawn uniformly from $[80, 600)$, used only for the token-share column of Table 6.

C.3 Fitting procedure

Parameters are fitted in two stages, for reasons given in Section 16. The first stage learns the weights by minimising the hinge objective with the closed-form gradients of Proposition 16.3: softmax parameterisation of w , softplus parameterisation of $p = 1 + \text{softplus}(a)$, learning rate 0.5 decaying linearly to 0.15 of its initial value, momentum 0.9, margin $\gamma = 0.10$, entropy regularisation 0.015, batch of 4 states per step, 48 sampled positives and negatives per state, 1200 iterations. The entropy term is not decoration: without it the softmax collapses onto a single view within a few hundred steps and the fitted geometry becomes a disguised single-view baseline.

The second stage selects the exponent on a held-out split (35% of states) by mean F_1 under the gap radius, over the grid $\{1, 1.5, 2, 3, 4, 6, 8, 16, \infty\}$. Selecting p by grid rather than by gradient is a deliberate concession to auditability. The exponent is a single scalar whose effect on the neighbourhood is highly non-linear near the crossover of Proposition 4.7, gradient descent on it is poorly conditioned in exactly that region, and a nine-point grid evaluated on held-out states costs less than the gradient path and yields a number an operator can defend. We regard this as the correct engineering choice and not merely an expedient one, though a practitioner who wants a differentiable end-to-end pipeline can substitute the smooth relaxation of Theorem 16.4 at the cost of that auditability.

C.4 Experiments and what each one is for

E0, gradient verification. Two hundred random configurations, central differences against the closed forms. This is a test of Proposition 16.3, and it is run first because every learned result downstream is worthless if it fails.

E1 and E1b, context quality. Ten methods on five seeds, 120 training and 250 test states. The baselines are semantic Top- k at four values of k , a hybrid semantic-plus-recency score, a one-hop graph expansion, a hand-written rule filter, and full exposure. E1b sweeps the decoy multiplier over $\{0, 1, 2, 4\}$.

E2, exponent behaviour. Numerical location of the crossover of Proposition 4.7 against its closed form, plus the neighbourhood-level consequence: Jaccard drift and pairwise rank-flip rate between $p = 1$ and $p = 8$.

E3, radius rules. The five rules of Section 7 on three seeds, scored by F_1 , neighbourhood size and exact-recovery rate.

E4, routing under policy. Six hundred tools, attenuation $a(k) = 1/(1+0.06k)$, sweeping ε to locate the optimum of Theorem 11.2 and to measure the survival of forbidden tools.

E5, planning. Forty random graphs, $|V| = 600$, degree 25 truncated to $k = 5$ by the neighbourhood, edge costs $\mathcal{U}[1, 2]$, comparing realised suboptimality against the bound of Theorem 13.4.

E6, ultrametric caching. Four thousand steps on a depth-4, branching-5 hierarchy with 85% local moves, measuring exact context reuse under an ultrametric geometry

against a flat one. This is the experiment that tests Theorem 5.9(iv), the sharpest practical claim in the paper.

E7, index filtering. Four hundred trials, $N = 4000$, checking the no-false-negative property of Theorem 17.2 and measuring the candidate fraction with and without a weight floor.

E8, concentration. Contrast ratios under uniform and sparse weights as the view count grows, testing Proposition 4.11.

E9, drift. Three hundred steps with two injected shocks replacing a quarter of the environment, measuring detection against a 3σ threshold calibrated on the quiet period, and the false-alarm rate.

E10, stability and constraints. Three hundred trials perturbing the weights by $\|\Delta w\|_1 = 0.05$, checking the bound of Theorem 6.5, the annulus containment of Theorem 6.7, and constraint enforcement under $p = \infty$.

C.5 Reproduction

Running `python experiments.py` regenerates `results.json` and the figures. Seeds are fixed per experiment and reported in the source; the tables in Section 19 are produced directly from `results.json` rather than transcribed, so a discrepancy between the paper and a rerun is a bug in one of the two and not a rounding difference.

D Notation

Symbol	Meaning
$\mathcal{X}, N$	object universe and its cardinality
M, K, T, S, A, C, H	memories, knowledge, tools, skills, agents, code, history
δ_i, n	the i -th view and the number of views
s, g	agent state and goal
p, w, ε	exponent, weights, radius
$d_{p,w}$	weighted L^p aggregate (Definition 4.1)
$D_\theta(s, x)$	state-anchored dissimilarity (Definition 6.1)
$N_\theta(s, g)$	adaptive neighbourhood, i.e. the context
$M_p(a)$	weighted power mean of the view vector a
n_{eff}	effective dimension $\ w\ _2^{-2}$
Dgm, d_B	persistence diagram, bottleneck distance
h^*, h_N	optimal and restricted cost-to-go
μ_τ	soft membership at temperature τ
θ	learned parameters

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